

DATA 423

Assignment

03

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Part A: Data Description and EDA

1). Introduce Data:

The dataset used in this data analysis is named *Ass3Data.csv*, which is an artificially generated dataset designed to simulate recorded health data of a particular response variable named *Response*. The dataset has a file size of approximately 323 KB and contains 969 rows of observations and 21 columns of variables, giving a total of 20,349 cells (rows × columns). Of these values, 19,770 entries are recorded without missingness, resulting in an overall data completeness of approximately 97.2%. There are 776 observations that are fully complete (80.1%), while the remaining 193 rows (19.9%) contain at least one missing value. In total, around 2.8% of all dataset cells contain missing values.

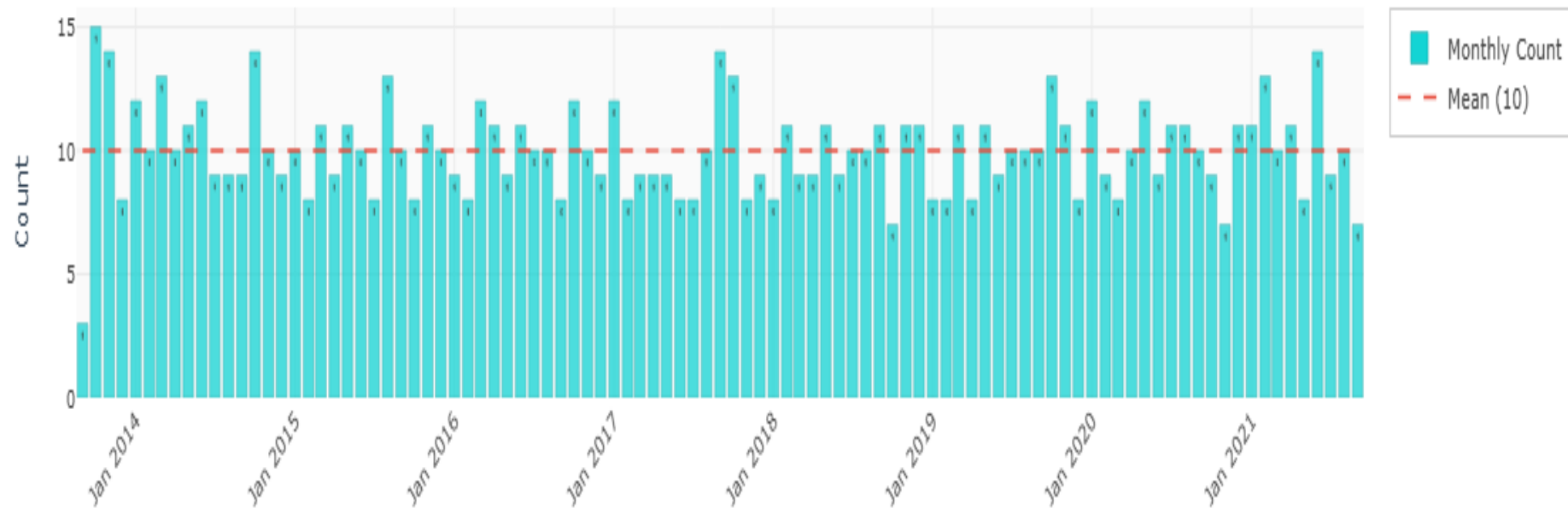
The *Patient* column acts as a unique identifier for each individual in the study. *Patient* identifiers range from values *tid-57095* to *tid-58063* with no skips or gaps in the sequence, and there are no missing values in this column. The other categorical variable is *BloodType*, which has a cardinality of four representing the ABO blood groups: *A*, *B*, *AB* and *O*. The blood groups are relatively equally distributed amongst patients, with *A* having 215 (22.2%), *B* having 228 (23.5%), *AB* having 286 (29.5%), and *O* having 240 (24.8%).

The *ObservationDate* specifies when the results were collected and uses a DD/MM/YYYY format. The dataset was already organised with the dates arranged in chronological order, spanning from 13th September 2013 to 21st September 2021. This gives a total timespan of 2,925 days, or 8.0 years. There are also no missing entries for *ObservationDate*. The most common year recorded is 2014 with 128 observations, representing 13.2% of the data. The most common month is October 2013 with 15 observations (1.5%), and the most common quarter is the fourth quarter of 2013, containing 37 observations (3.8%). A distribution of the number of entries per month from this 8-year period can be seen in **Visualisation 01** and was relatively consistent across the 8-year period with a mean of 10 observations per month.

Within the dataset, there are 19 numeric variables, making up 90.5% of all variables. These include *Alcohol*, *Coffee*, *Exercise*, *ChemoTreatments*, *ReagentA* through *ReagentN*, and the primary outcome variable *Response*. These numeric variables represent a mixture of lifestyle and biomarker related measurements for the respective patients in the trial. The response variable of primary interest in this analysis is *Response*, which is assumed to be a measurable biomarker of interest in this study. *Response* contains no missing entries. There are also no missing values for the lifestyle factor variables of *Alcohol*, *Coffee*, *Exercise*, or *ChemoTreatments*. Missing entries are only present among the reagent variables, ranging from 33 missing values for *ReagentH* to 52 missing values for *ReagentN*. All the missingness from the dataset comes from these variables, where they are denoted with NULL values.

Examining the numeric summaries presented in **Table 1**, the lifestyle variables exhibit relatively limited ranges. This is evident in the following: *Alcohol* ranges from 0.00 to 4.00 (mean = 2.01), *Coffee* from 0.00 to 7.00 (mean = 3.52), *Exercise* from 0.00 to 11.99 (mean = 6.06), and *ChemoTreatments* from 0.00 to 4.99 (mean = 2.48). The reagent variables demonstrate broadly similar distributions, with mean values generally lying between 500 and 700 for most reagents, with the exception of *ReagentE* and *ReagentM*, which possess lower means of approximately 200. The *Response* variable, which constitutes the primary focus of the present analysis, ranges from -375.28 to 6,250.15, with a mean of 2,999.96 and a standard deviation of 916.58.

Monthly Distribution — ObservationDate



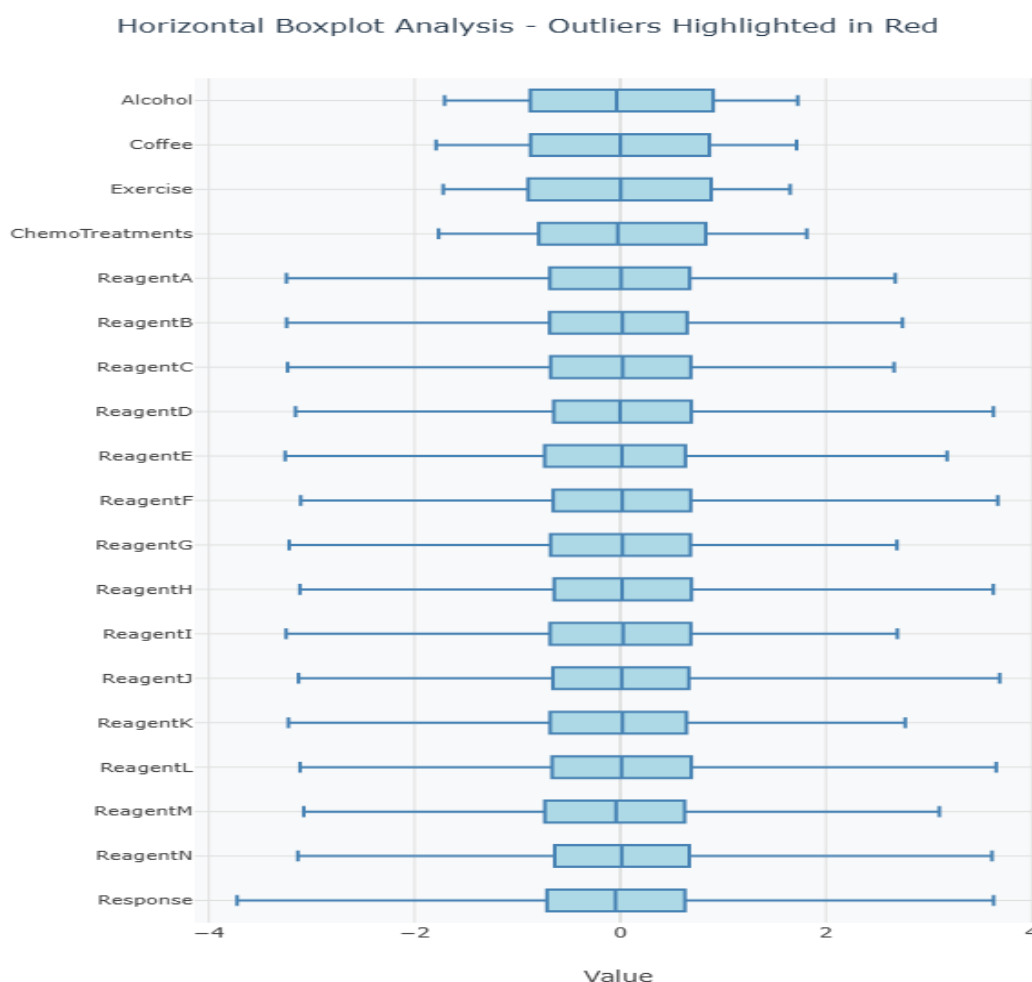
Visualisation 01: Bar graph showing the monthly distribution of recorded observations between 13th September 2013 and 21st September 2021. The distribution of entries is relatively consistent across the 8-year period, with an average of approximately 10 observations recorded per month and no substantial gaps or periods of unusually high concentration.

Table 01: Numeric Variable Distribution and Missingness Summary										
Variable	Min	Q1	Median	Mean	Q3	Max	SD	N	Missing	Missing %
Alcohol	0	1.01	1.97	2.01	3.06	4	1.16	969	0	0%
Coffee	0	1.8	3.55	3.52	5.26	7	2	969	0	0%
Exercise	0	2.9	6.17	6.06	9.15	11.99	3.52	969	0	0%
ChemoTreatments	0	1.31	2.44	2.48	3.69	4.99	1.41	969	0	0%
ReagentA	301.19	613.56	695.8	695.57	776.26	1,020.06	120.83	935	34	3.50%
ReagentB	230.42	442.14	501.19	499.74	554.7	725.65	83.35	926	43	4.40%
ReagentC	302.8	615.7	698.19	696.88	778.64	1,018.50	121	930	39	4%
ReagentD	283.28	534	601.21	600.02	667.53	962.98	100.76	926	43	4.40%
ReagentE	71.73	172.5	201.79	201.36	225.58	326.41	39.4	926	43	4.40%
ReagentF	288.91	534.39	602.21	600.2	667.14	966.64	100.71	927	42	4.30%
ReagentG	304.14	618.74	700.04	699.25	779.98	1,022.25	120.57	928	41	4.20%
ReagentH	287.63	535.88	603.07	600.95	668.84	963.11	100.71	936	33	3.40%
ReagentI	300.51	616.32	700.07	697.58	779.9	1,022.77	121.15	935	34	3.50%
ReagentJ	285.42	533.81	603.05	599.54	665.8	969.72	99.81	922	47	4.90%
ReagentK	230.96	439.82	501.12	499.13	553.46	728.64	83.75	918	51	5.30%
ReagentL	287.3	532.57	603.11	599.84	666.16	965.98	99.99	934	35	3.60%
ReagentM	79.96	171.97	198.97	200.49	225.08	322.47	39.59	927	42	4.30%
ReagentN	285.77	534.94	602.47	599.78	665.01	961.71	100.52	917	52	5.40%
Response	-375.28	2,366.64	2,948.10	2,999.96	3,565.00	6,250.15	916.58	969	0	0%

Table 01: Summary statistics and missing value counts for all numeric variables in the dataset, including lifestyle variables, reagent measurements, and the *Response* variable. The table shows that most reagent variables have similar distributions, while *Response* exhibits the largest variability. Missing values are present only within the reagent variables, whereas *Alcohol*, *Coffee*, *Exercise*, *ChemoTreatments*, and *Response* contain complete observations.

2). Boxplot:

Boxplots were constructed for all numeric variables using the standard 1.5 IQR rule to identify potential outliers. No outliers were detected for the lifestyle variables *Alcohol*, *Coffee*, *Exercise*, and *ChemoTreatments*, indicating relatively compact distributions without extreme values. Among the reagent variables, a small number of outliers were identified, ranging from 2 outliers for Reagents A, C, G, and I to 9 outliers for Reagents J and N. Most of these outliers fell below the lower bound of the boxplots. The *Response* variable contained 10 outliers, with 9 above the upper bound and 1 below the lower bound. When the IQR multiplier was increased from 1.5 to 2.3, all detected outliers disappeared across every variable. This suggests that the previously identified outliers were not sufficiently extreme to be considered true anomalies and are more likely to represent the natural tails of the distributions rather than data entry or measurement errors. **Visualisation 02** presents a centred and scaled boxplot of the numeric variables using an IQR multiplier of 2.3. The visualisation clearly shows that *Alcohol*, *Coffee*, *Exercise*, and *ChemoTreatments* exhibit substantially lower variability and narrower value ranges compared with the reagent variables and *Response*.



Visualisation 02: Centred and scaled boxplots of all numeric variables using an IQR multiplier of 2.3. The lifestyle variables (*Alcohol*, *Coffee*, *Exercise*, and *ChemoTreatments*) display comparatively low variability and narrower distributions, while the reagent variables and *Response* exhibit greater spread and variability across observations.

3). Correlation:

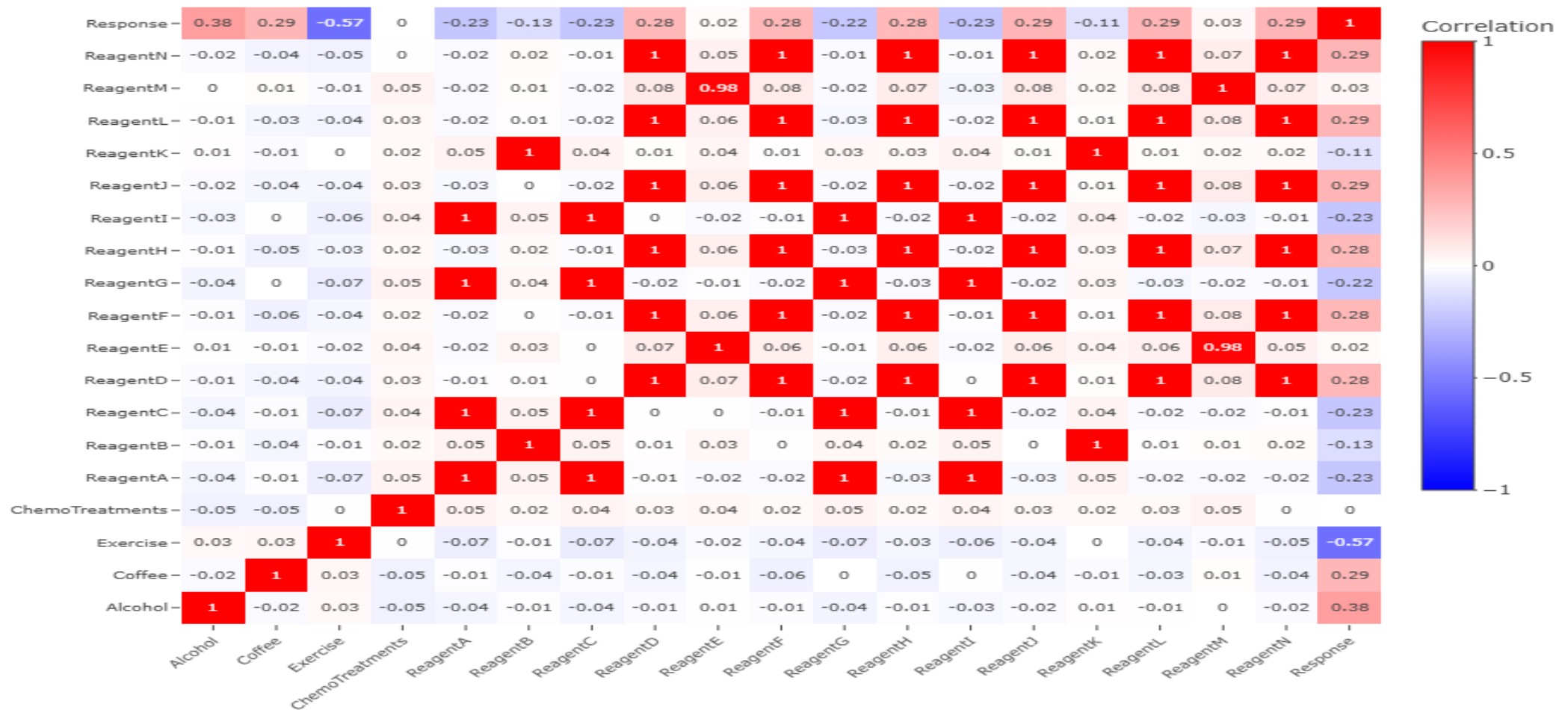
The Pearson correlation heatmap in **Visualisation 03** shows the linear relationships between all numeric variables in the dataset, including observations containing missing reagent values. Most variables exhibit weak correlations, with coefficients generally close to zero, indicating limited linear association between the majority of predictors.

Among the lifestyle variables, *Exercise* shows the strongest relationship with *Response*, with a moderate negative correlation of $r = -0.57$, suggesting that higher exercise levels are associated with lower response values. *Alcohol* and *Coffee* both show weak positive correlations with *Response* ($r = 0.38$ and $r = 0.29$, respectively), while *ChemoTreatments* has virtually no correlation with *Response* ($r \approx 0.00$).

The reagent variables display several very strong positive correlations with one another, with many pairs approaching $r = 1.00$. This indicates that several reagents may measure highly related biological processes or contain substantial multicollinearity. In contrast, correlations between most reagent variables and the *Response* variable are relatively weak, typically ranging between approximately -0.23 and 0.29 .

Overall, the heatmap suggests that *Exercise* has the clearest linear relationship with the response variable, while the reagent variables appear highly intercorrelated with each other. This pattern may indicate redundancy among some reagent measurements and suggests that dimensionality reduction or feature selection techniques may be useful in subsequent modelling stages.

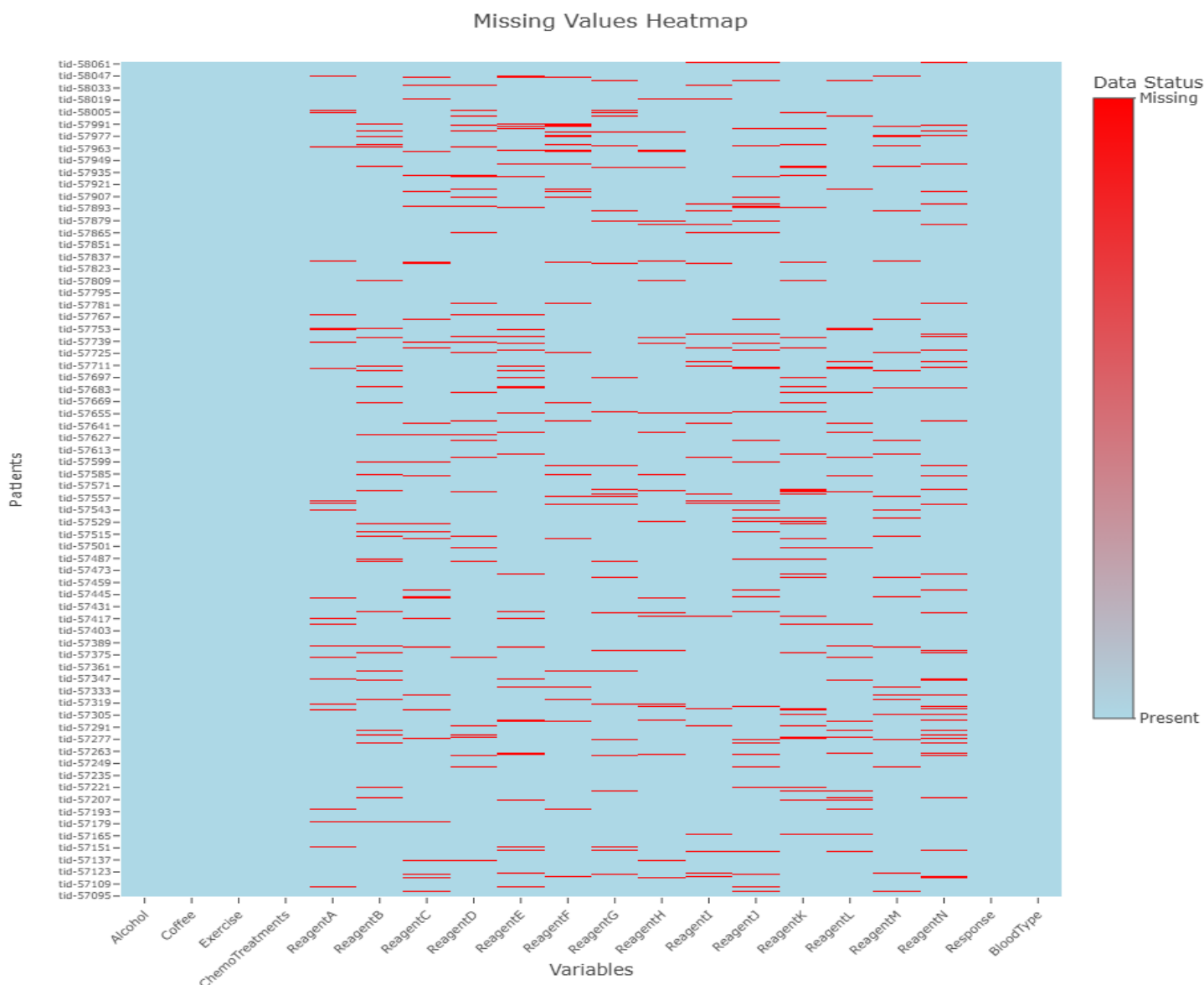
PEARSON Correlation Matrix



Visualisation 03: Pearson correlation heatmap for all numeric variables, including observations containing missing reagent entries. The heatmap shows generally weak correlations between most variables, with *Exercise* displaying the strongest relationship with *Response* ($r=-0.57$). Several reagent variables exhibit extremely strong positive correlations with one another, indicating substantial multicollinearity. This suggests that some reagents may represent closely related biological measurements to one another.

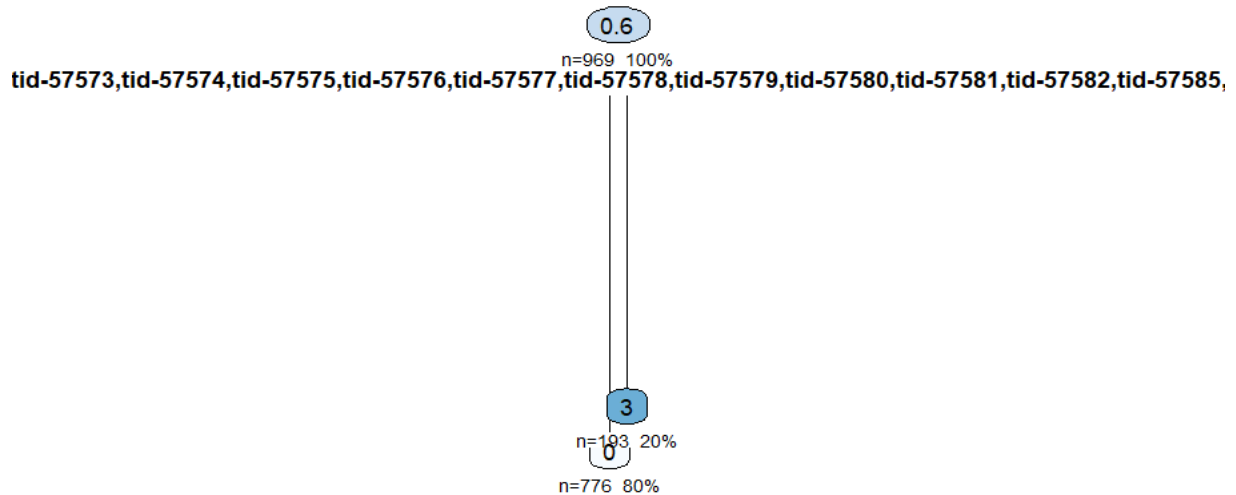
4). Missingness Heatmap and Tree:

As discussed in the Introduce Data section, the missingness heatmap in **Visualisation 04** shows missing NULL entries only within the reagent variables, ranging from 3.4% missing values for *ReagentH* to 5.3% for *ReagentN*. However, further investigation using a decision tree model as seen **Visualisation 05** to predict missingness per observation revealed a clear non-random pattern: missing values are not dispersed randomly throughout the dataset. Instead, 193 specific patients (19.9% of the sample) each have exactly 3 missing values, while the remaining patients have complete data. This indicates the missingness is structured by patient identifier, suggesting a systematic issue affecting particular individuals rather than random occurrence across the dataset.



Visualisation 04: Missingness heatmap showing the distribution of NULL values across the reagent variables. Missing values are limited to the reagent measurements, ranging from 3.4% (*ReagentH*) to 5.3% (*ReagentN*) of observations. While the heatmap suggests dispersion throughout the dataset, further analysis using a decision tree (**Visualisation 05**) reveals that missingness is not random but systematically concentrated within a specific subset of 193 patients, each having exactly three missing reagent values. The remaining 776 patients have complete data.

Predicting Missingness per Observation



Visualisation 05: Decision tree predicting the number of missing values per observation. The tree splits exclusively on the *Patient* variable, separating 969 observations into two groups: 776 patients with 0 missing values (complete data) and 193 patients with exactly 3 missing reagent values. The root node displays the overall mean missingness of 0.6. The absence of any other splitting variables confirms that patient identity alone predicts missingness, rejecting MCAR and indicating a structured non-random missingness pattern.

Part B: Strategies

1). Missing Data:

There were no excessively missing variables or observations that required removal from the dataset. Missing values were limited to the reagent variables only, with relatively low levels of missingness ranging from 3.4% for ReagentH to 5.3% for ReagentN.

However, further investigation showed that the missingness pattern was highly structured rather than random. A decision tree analysis (**Visualisation 05**) predicting the number of missing values per observation showed that missingness was almost entirely explained by the *Patient* variable (variable importance = 1391), whereas all other variables had minimal importance (largest remaining importance = 7.2). The tree produced a perfect split between 776 patients with no missing values and 193 patients with exactly three missing values.

This provides strong evidence that the missing data mechanism is not Missing Completely at Random (MCAR). Instead, the strong dependence on patient identifier suggests a Missing Not at Random (MNAR) pattern, where the probability of missingness is related to underlying patient-specific characteristics.

Because of this structured missingness pattern, deletion-based approaches were considered inappropriate, as they would likely introduce substantial bias. Consequently, imputation was considered the most appropriate preprocessing strategy. Given the moderate observation-to-variable ratio and the clustered missingness structure, k-nearest neighbours (k-NN) imputation with k=5 was selected. This approach was appropriate because it preserves multivariate relationships between reagent variables, uses information from the 776 complete patient records to estimate missing values for affected patients, and reduces the risk of bias associated with the non-random missingness pattern.

Since the *Response* variable contained no missing values, no target observations required removal. The k-NN imputation procedure was applied consistently across all candidate modelling workflows.

2). Outliers:

Outlier analysis using the standard 1.5 IQR rule identified only a small number of mild outliers across several reagent variables and the *Response* variable. However, when the IQR multiplier was increased to 2.3, all detected outliers disappeared, indicating that the extreme values are more likely to represent the natural tails of the distributions rather than genuine anomalies or data quality issues. As a result, no observations were removed from the dataset, and no specialised robust outlier treatment methods were considered necessary.

3). Feature Engineering:

Several preprocessing and feature engineering steps were incorporated into a common modelling recipe applied across all candidate methods, ensuring fair and consistent comparison.

Missing predictor values were first handled using k-NN imputation (k=5). Date variables were then decomposed into day-of-week, which was selected over alternative representations (decimal date, month) based on comparative performance during initial preprocessing experiments, suggesting a within-week temporal pattern in the data. Categorical variables were encoded using dummy encoding due to their low cardinality; specifically, *BloodType* (4 categories) and day-of-week (7 categories).

Zero variance (ZV) and near-zero-variance (NZV) were subsequently applied to remove predictors whose near-constant values risk causing instability during model fitting and cross-validation. Finally, centring and scaling were applied to standardise the predictor space across the differing scales and variances present in the numeric variables.

Pearson correlation analysis identified strong correlations among several reagent variables, indicating multicollinearity. Dimensionality reduction via principal component analysis (PCA) was not applied during the main modelling phase, as the author was not sufficiently familiar with the method at that time. In retrospect, techniques like PCA could have been explored, but the models ultimately performed adequately without it. This is acknowledged as a limitation of the current analysis.

4). Recipe:

The chosen recipe that was applied to all candidate methods followed:

Step 1: impute_knn (k = 5)

Step 2: dow

Step 3: dummy

Step 4: zv

Step 5: nzv

Step 6: center

Step 7: scale

This recipe was chosen to produce a baseline result and assess the general performance of each model before any model-specific tuning or further feature refinement.

5). Tuning and Assessing:

A static test set, consisting of 20% of the observations, was reserved for final model assessment. This partition was created using stratified sampling on the *Response* variable, with a fixed random seed of 199. The remaining 80% of observations were allocated to model development and hyperparameter tuning.

Hyperparameter optimisation was conducted using 25 bootstrap resamples of the training data, employing a grid search strategy across candidate hyperparameter values. To ensure full reproducibility of the modelling workflow and results, fixed random seeds were applied throughout: seed 199 for the data partition and seed 673 for the generation of resampling indices.

Part C: Methods

1). Candidate Methods:

Candidate methods were selected to ensure broad coverage across distinct regression modelling families, rather than to maximise the number of methods evaluated. The primary criterion for inclusion was that each method represented a meaningfully different approach to learning from data, whether through different assumptions about the underlying data generating process, different optimisation strategies, or fundamentally different predictive objectives. This allowed the study to compare linear, nonlinear, parametric, nonparametric, Bayesian, kernel-based, ensemble, and instance-based methods as examples within a unified caret framework. Approximately 30 candidate methods were targeted to enable a sufficiently broad search of potential regression methodologies. A summary of each candidate method, including model family, characteristic notes, and rationale for inclusion, is provided in **Table 02**.

A secondary consideration was the availability of tuneable hyperparameters, with methods preferred where tuning could be applied in a meaningful way. To support the selection process and provide a visual representation of how the candidate methods relate to one another, method similarity was assessed using multidimensional scaling (MDS), as shown in **Visualisation 06**. Pairwise distances between methods were computed across their full set of hyperparameter and performance characteristics using Euclidean distance via `stats::dist`, and then projected onto two dimensions using classical MDS. This allowed the 27 successfully trained models to be positioned relative to the broader pool of available caret methods, confirming that the chosen set achieved meaningful spread across the method space rather than clustering within a narrow region. Methods shown in blue were retained for analysis; those shown in red (gbm, xgbTree, M5, and icr) were excluded due to software incompatibilities or instability within the caret workflow.

Table 02 (See next two pages): Summarises the 27 candidate regression methods evaluated within the caret framework. For each method, the table reports the model family, key characteristics, relevant implementation notes, and the rationale for inclusion. The candidate set was deliberately selected to ensure broad coverage across distinct modelling families including linear, nonlinear, parametric, nonparametric, Bayesian, kernel based, ensemble, and instance-based approaches, rather than to maximise the total number of methods. Methods that failed during implementation, such as gbm, xgbTree, M5, and icr, are highlighted in red at the bottom of the table with the specific technical reason for each failure provided.

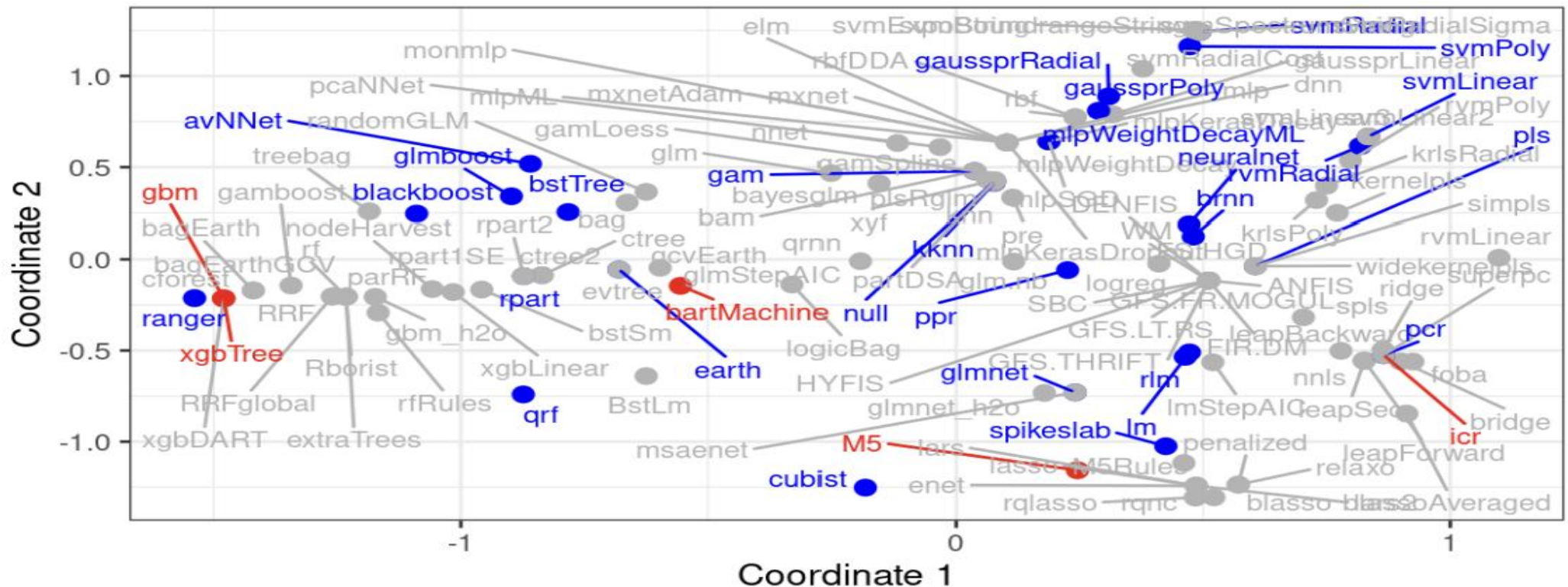
Table 02: Method Summary Table

Method #:	Method Name:	Model Family:	Characteristics:	Notes:	Reason Selected:
1	lm - Linear Regression	Linear Model	Linear Model. Parametric. No regularisation	No hyperparameters	Baseline benchmark for all other models
2	glmnet - Elastic Net	Linear Model	Linear Model. Regularised. Variable Selection	alpha, lambda	Combines ridge and lasso regularisation in a single framework
3	rlm - Robust Linear Model	Linear Model	Linear Model. Robust to outliers	Weighting function	Reduces sensitivity to extreme values and influential observations
4	pls - Partial Least Squares	Latent Component	Latent Component Model. Dimensionality Reduction	ncomp. tuneLength 25	Maximises covariance between predictors and response
5	pcr - Principal Component Regression	Latent Component	Latent Component Model. Dimensionality Reduction	ncomp. tuneLength 25	Contrasting approach to PLS using predictor variance alone
6	rpart - CART Decision Tree	Tree Based Mode	Tree Based Model. Nonlinear. Interpretable	cp	Captures nonlinear relationships through recursive partitioning
7	cubist - Cubist	Tree Based Mode	Rule Based Model. Linear models at terminal nodes. Neighbour adjustment	committees, neighbors	Hybrid approach combining rules with instance-based adjustment
8	ranger - Random Forest	Random Forest	Ensemble Model. Bagging. Permutation Importance	mtry, min.node.size, splitrule. 500 trees	Strong nonlinear learner with low overfitting via bootstrap aggregation
9	qrf - Quantile Random Forest	Random Forest	Ensemble Model. Bagging. Quantile Estimation	mtry	Estimates prediction intervals rather than conditional means only
10	bstTree - Boosted Trees	Gradient Boosting	Ensemble Model. Sequential Boosting. Tree Base Learner	mstop, maxdepth, nu	Iterative weak learner improvement on structured tabular data
11	glmboost - Boosted Linear Model	Gradient Boosting	Ensemble Model. Sequential Boosting. Linear Base Learner	mstop. Values 50, 100, 200, 500	Sparse additive boosting framework using linear base learners
12	blackboost - Black Box Boosting	Gradient Boosting	Ensemble Model. Sequential Boosting. Conditional Inference Trees	mstop, maxdepth	Statistically different splitting strategy to conventional boosting
13	svmRadial - SVM Radial Kernel	SVM	Kernel Method. Nonlinear. Maximum Margin	C, sigma	Flexible nonlinear modelling capturing complex local relationships
14	svmPoly - SVM Polynomial Kernel	SVM	Kernel Method. Nonlinear. Maximum Margin	degree, scale, C	Captures nonlinear feature interactions through polynomial transformations
15	svmLinear - SVM Linear Kernel	SVM	Kernel Method. Linear. Maximum Margin	C	Approximates linear relationships within a kernel framework
16	avNNet - Model Averaged Neural Network	Neural Network	Neural Network. Averaged Ensemble. Single Hidden Layer	size, decay	Reduces instability from random weight initialisation by averaging across networks

17	mlpWeightDecay ML - Multi Layer Perceptron	Neural Network	Neural Network. Deep Architecture. Weight Decay Regularisation	hidden layer architecture, decay	Allows tuning of deeper architectures with regularisation control
18	brnn - Bayesian Regularised Neural Network	Neural Network	Neural Network. Bayesian Regularisation	neurons	Reduces overfitting without requiring extensive manual regularisation tuning
19	neuralnet - Neural Network Backpropagation	Neural Network	Neural Network. Feed Forward. Flexible Hidden Layer Structure	layer1, layer2, layer3	Traditional implementation with flexible hidden layer and activation behaviour
20	gaussprRadial - Gaussian Process Radial	Gaussian Process	Gaussian Process. Kernel Method. Probabilistic	sigma	Probabilistic nonlinear modelling with uncertainty quantification
21	gaussprPoly - Gaussian Process Polynomial	Gaussian Process	Gaussian Process. Kernel Method. Probabilistic	degree, scale	Different smoothness assumptions to gaussprRadial via polynomial kernel
22	earth - MARS	GAM/MARS	Nonlinear Regression. Adaptive Piecewise Linear. Automatic Interaction Detection	degree, nprune	Models nonlinear relationships with automatic interaction detection
23	gam - Generalised Additive Model	GAM/MARS	Nonlinear Regression. Spline Based. Additive	select, method	Interpretable nonlinear modelling through smooth additive components
24	ppr - Projection Pursuit Regression	GAM/MARS	Nonlinear Regression. Learned Projections	nterms	Fundamentally different nonlinear approach based on learned projection directions
25	spikeslab - Spike and Slab Regression	Bayesian Model	Bayesian Model. Sparse. Variable Selection	vars	Probabilistic variable selection fundamentally different to penalised regression
26	rvmRadial - Relevance Vector Machine	Bayesian Model	Sparse Bayesian Kernel Method	sigma	Produces compact models with fewer support vectors than SVM
27	kkn - Weighted K Nearest Neighbours	K Nearest Neighbour	Instance Based Learning. Distance Weighted	k	Flexible instance-based learning with kernel based neighbour contributions
NA	gbm - Gradient Boosting Machine	Gradient Boosting	Ensemble Model. Sequential Boosting. Tree Base Learner	n.trees, interaction.depth, shrinkage, n.minobsinnode	Failed. Persistent incompatibility with recipes preprocessing pipeline
NA	xgbTree - XGBoost	Gradient Boosting	Ensemble Model. Sequential Boosting. Regularised Trees	nrounds, max_depth, eta, gamma, colsample_bytree, min_child_weight, subsample	Failed. Strict matrix input requirements caused dimension mismatches with recipes
NA	M5 - M5 Model Trees	Tree Based Mode	Tree Based Model. Linear Models at Terminal Nodes	pruned, smoothed, rules	Failed. RWeka Java integration proved unreliable within the project environment
NA	icr - Independent Component Regression	Latent Component	Latent Component Model. Statistical Independence	n.comp	Failed. fastICA implementation unstable due to strict numeric matrix requirements

Regression Methods

Model Status ● Working Models ● Failed Models



Visualisation 06: Two-dimensional multidimensional scaling (MDS) plot of regression method similarity, where each point represents a candidate caret regression method positioned according to its similarity to all other methods in the space. Pairwise distances were computed across model characteristics using Euclidean distance via `stats::dist` and projected onto two coordinates using classical MDS scaling, such that methods appearing closer together share greater similarity in their modelling approach, while methods positioned further apart represent more distinct approaches.

Working models retained for analysis are highlighted in blue, while models that failed during training or were excluded due to software incompatibilities are highlighted in red. The grey points represent the broader pool of available caret regression methods that were considered but not selected. The working methods are spread reasonably well across the plot, indicating that the selection covers a broad range of different regression approaches rather than focusing on any one methodological area.

2). Candidate Models:

Table 03 presents a summary of cross-validated performance results across 27 candidate models (excluding the null model), ranked by Root Mean Square Error (RMSE). To complement this, the distribution of these results across resampled trials is shown in **Visualisation 07**, which plots MAE, RMSE, and R^2 scaled relative to the null model. While most models displayed relatively stable resampled distributions, a few of methods (notably ppr and gam) exhibited visible outliers across resampled folds, suggesting occasional instability in their predictions depending on the composition of the training data. The top performing models, by contrast, showed tighter distributions with few extreme resampled values, indicating more consistent and reliable behaviour across folds.

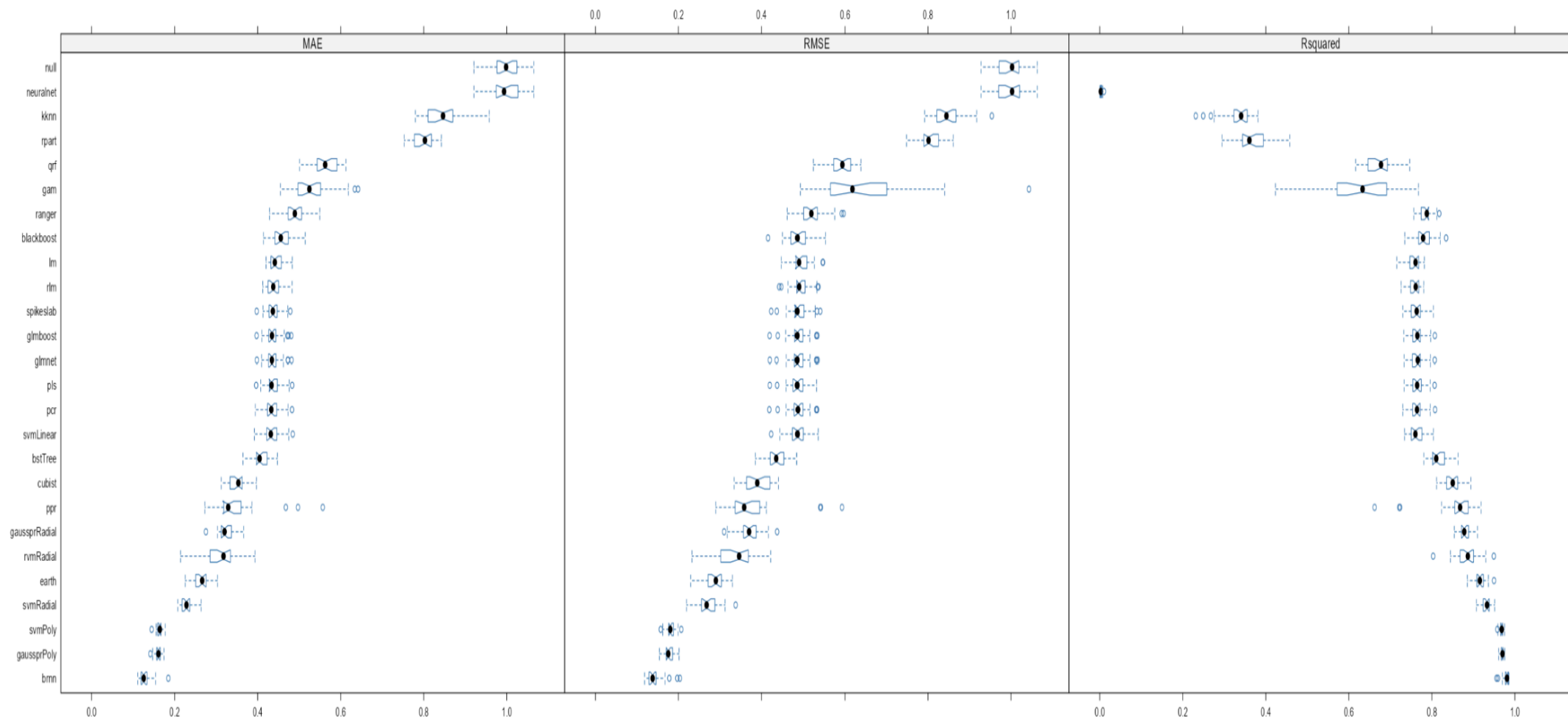
A clear performance hierarchy emerges from the results. The Bayesian Regularised Neural Network (brnn) achieved the strongest predictive performance by a considerable margin, returning an RMSE of 132.9, MAE of 93.3, and an R^2 of 0.979, indicating that the fitted model accounts for approximately 97.9% of variance in the outcome. The Gaussian Process with Polynomial kernel (gaussprPoly) and Support Vector Machine with Polynomial kernel (svmPoly) ranked second and third respectively, with near-identical R^2 values of around 0.969 and 0.968, though both trailed brnn by roughly 30 RMSE units. A middle tier of methods, including svmRadial, earth, and cubist, produced moderate performance with R^2 values between 0.85 and 0.93. Notably, a cluster of predominantly linear methods including pls, pcr, glmnet, lm, and rlm converged on virtually identical performance at $RMSE \approx 449\text{--}457$ and $R^2 \approx 0.76$, suggesting these approaches share a common performance ceiling likely attributable to the nonlinear structure of the data. At the lower end of the table, rpart and kkn performed poorly, while neuralnet performed no better than the null model. The avNNet and mlpWeightDecayML configurations produced errors substantially exceeding the null baseline, indicating model instability or misconfiguration.

Training time across candidate methods is summarised in **Visualisation 08**. The majority of models trained within approximately 15 seconds to three minutes, making computational cost a minor consideration for most methods. Notable exceptions were gam, neuralnet, and rvmRadial, which required 5 minutes 16 seconds, 6 minutes 45 seconds, and 12 minutes 34 seconds respectively, considerably longer than the remainder of the candidate set. Given that rvmRadial and neuralnet also produced comparatively poor predictive performance, their elevated training cost offers no compensating benefit.

On the basis of these results, brnn was selected as the preferred model, offering the best balance of accuracy, consistency, and training efficiency.

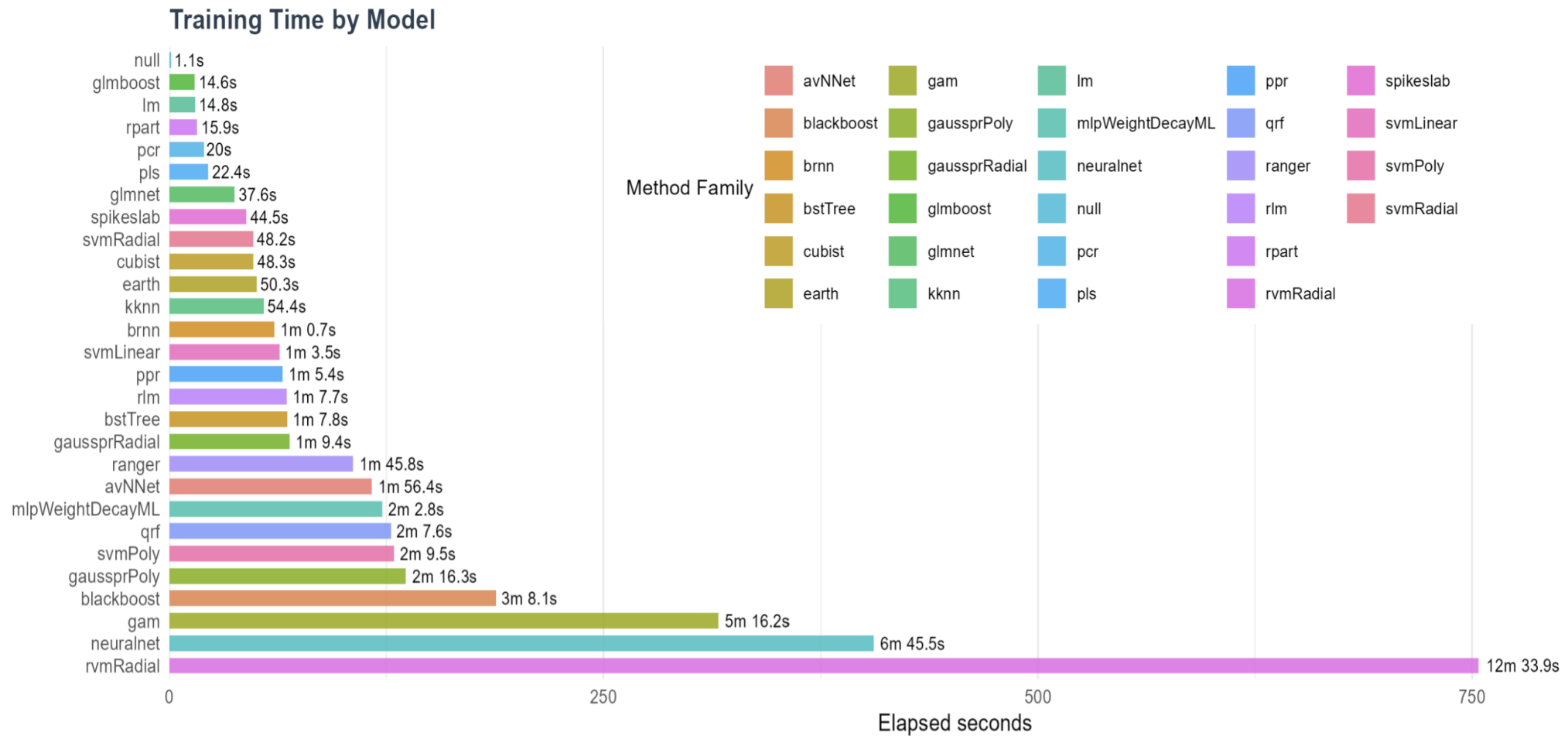
Method	RMSE	RMSESD	R ²	R ² SD	MAE	MAESD	BestTune	Train Time
brnn	132.877	20.8331	0.9788	0.008	93.2786	11.6546	neurons=4	1m 0.7s
gaussprPoly	163.988	10.7008	0.9688	0.0039	116.0855	5.848	degree=2, scale=0.1	2m 16.3s
svmPoly	167.6164	10.8598	0.9675	0.0047	118.122	4.7254	degree=2, scale=0.1, C=0.25	2m 9.5s
svmRadial	249.5483	24.3753	0.9311	0.0118	165.2923	10.0782	sigma=0.0239, C=4	48.2s
earth	266.5679	21.6372	0.917	0.0131	190.7113	13.4242	nprune=20, degree=2	50.3s
rvmRadial	311.5992	43.2779	0.8858	0.0305	226.85	27.6946	sigma=0.001	12m 33.9s
gaussprRadial	344.6035	27.909	0.879	0.015	234.2836	14.3975	sigma=0.01	1m 9.4s
ppr	350.3723	70.0498	0.8547	0.0626	253.8538	49.1747	nterms=3	1m 5.4s
cubist	361.6578	28.6356	0.8494	0.022	253.8049	18.2752	committees=20, neighbors=5	48.3s
bstTree	403.0654	23.3853	0.8173	0.0203	294.2868	15.0835	mstop=200, maxdepth=3, nu=0.1	1m 7.8s
blackboost	448.2864	27.0808	0.7819	0.024	333.2151	18.4458	mstop=100, maxdepth=3	3m 8.1s
pls	448.7685	23.7619	0.7649	0.0178	316.7074	15.4633	ncomp=5	22.4s
pcr	448.9288	24.1817	0.7648	0.0178	315.6001	15.2669	ncomp=16	20s
glmnet	449.3128	24.0479	0.7648	0.017	316.6128	14.745	alpha=0.55, lambda=10.4413	37.6s
glmboost	449.412	23.9012	0.7645	0.0171	316.5653	14.7843	mstop=500, prune=no	14.6s
spikeslab	449.7182	23.9525	0.764	0.0173	317.2944	14.6769	vars=27	44.5s
svmLinear	450.7279	23.7402	0.7632	0.0168	314.5001	15.3074	C=0.1	1m 3.5s
rlm	454.8391	22.3557	0.7589	0.0151	318.191	14.1551	intercept=TRUE, psi=psi.huber	37.7s
lm	456.6368	21.7567	0.7569	0.0156	323.0578	13.2739	intercept=TRUE	14.8s
ranger	481.2945	30.0347	0.7847	0.0153	354.2372	22.0744	mtry=20, splitrule=extratrees, min.node.size=5	1m 45.8s
qrf	544.6737	28.5387	0.6748	0.0321	409.3043	20.8834	mtry=14	2m 7.6s
gam	603.4554	116.2609	0.6212	0.0865	386.1645	37.3678	select=TRUE, method=GCV.Cp	5m 16.2s
rpart	741.4344	29.144	0.3699	0.0381	577.411	19.1453	cp=0.0257	15.9s
kknn	782.9667	35.8327	0.3331	0.0397	612.033	31.9102	kmax=13, distance=2, kernel=optimal	54.4s
neuralnet	922.8778	34.3795	0.0037	0.0028	723.2209	27.3333	layer1=5, layer2=0, layer3=0	6m 45.5s
null	923.1446	32.9413	N/A	N/A	723.8446	26.2989	parameter=none	1.1s
mlpWeightDecayML	1,374.25	267.9432	0.0493	0.0422	1,101.54	215.2083	layer1=7, layer2=0, layer3=0, decay=0.001	2m 2.8s
avNNet	3,147.80	35.5587	N/A	N/A	3,010.75	34.1256	size=1, decay=0.01, bag=FALSE	1m 56.4s

Table 03: Summary metric table of candidate models. A cross-validated performance metrics for 27 candidate models (excluding null model), including RMSE, MAE, and R², reported alongside their standard deviations across resampled folds. Models are ranked by RMSE from best to worst performing, with the null model (highlighted in green) included as a baseline reference. The best performing candidate model was brnn with 133 RSME, followed by gaussprPoly, svmPoly, and svmRadial. These top four performing models are highlighted in yellow.



Visualisation 07: Candidate model performance across resampled trials. Boxplots of cross-validated MAE, RMSE, and R^2 for all candidate models, scaled relative to the null model baseline. Each box summarises the distribution of metric values across resampled folds, with outlying points indicating occasional variation in model performance depending on training data composition. Two models of `mlpWeightDecayML` and `avNNet` were removed from the visualisation due to performing worse than the null model. The best performing candidate model was `brnn` with an RSME of 133.

Training Time Comparison



Visualisation 08: Bar chart of training time (in elapsed seconds) for each candidate model. The majority of methods trained within a range spanning approximately 15 seconds to 3 minutes. Only a few notable exceptions were observed, namely gam, neuralnet, and rvmRadial, which required substantially longer compute times for training.

3). Model Optimisation:

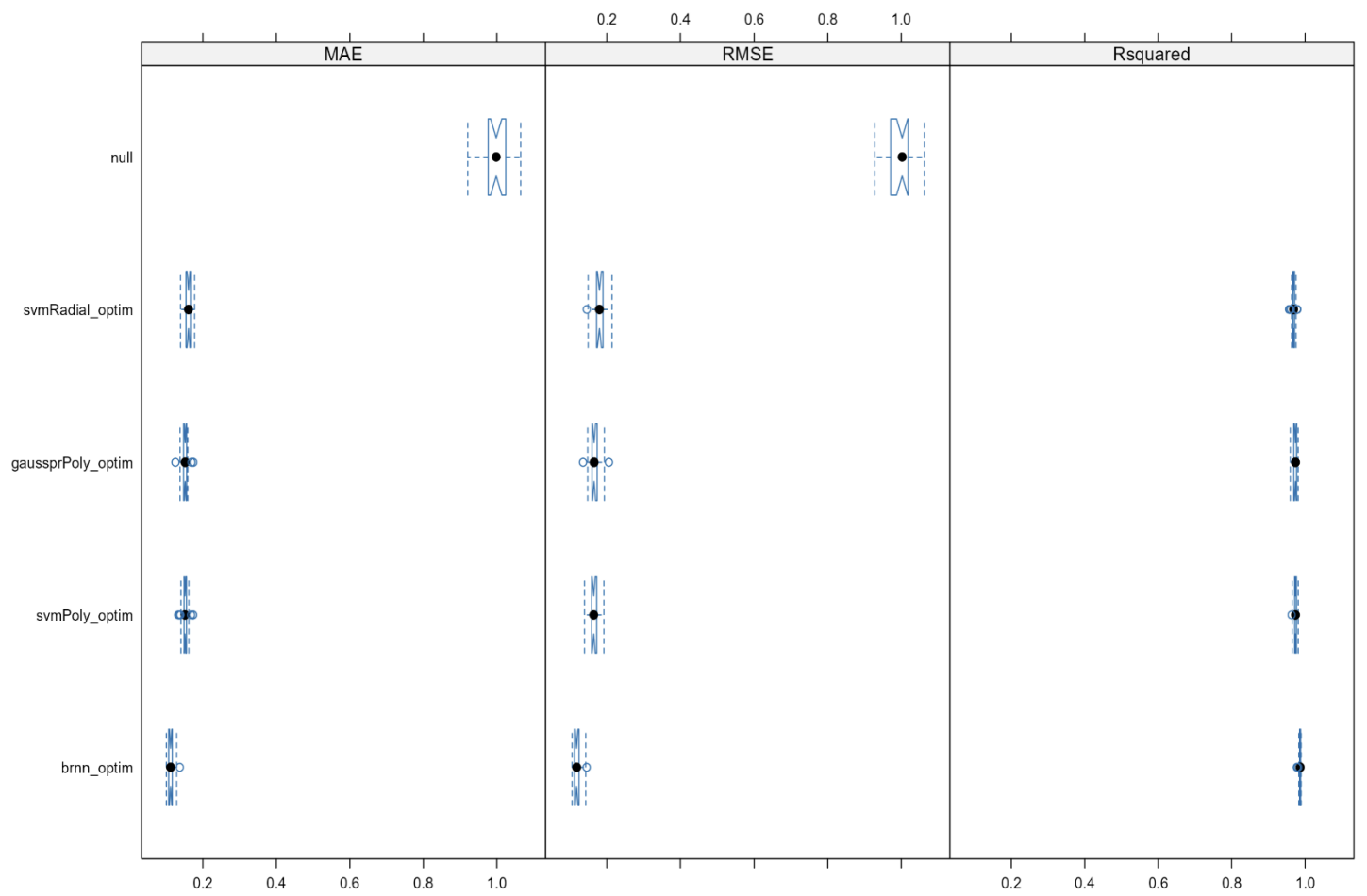
Following the initial candidate evaluation, the four best performing methods were selected for further refinement through expanded hyperparameter optimisation. In addition, the imputation strategy was revised at this stage; bagging imputation (using 25 trees) was found to produce marginally improved results compared to the previously used k-nearest neighbour imputation and was adopted for all optimised models. The hyperparameter search grids were also substantially expanded relative to the initial candidate run. For brnn, the neuron range was extended from the default to 1 to 10. For gaussprPoly, degree was searched across 1 to 5 and scale across 0.01 to 1. For svmPoly, degree was searched across 1 to 4, scale across 0.001 to 1, and cost across 0.1 to 20. For svmRadial, sigma was searched across 0.001 to 1 and cost across 0.1 to 50. The resampled distributions of MAE, RMSE, and R^2 for all optimised models, scaled relative to the null model baseline, are shown in **Visualisation 08**.

Table 04 presents the cross-validated performance results for the four optimised models. brnn remained the strongest performer following optimisation, achieving an RMSE of 110.5, MAE of 82.2, and an R^2 of 0.986, representing a meaningful improvement over its initial candidate result of RMSE 132.9 and R^2 of 0.979. Notably, brnn achieved this with a comparatively simple search space, tuning only a single hyperparameter across ten values, whereas the other methods required exploration across substantially larger multi-dimensional grids. svmPoly improved from an initial RMSE of 167.6 to 152.4, and gaussprPoly improved from 164.0 to 154.8, with both methods converging on similar performance despite their expanded search spaces. svmRadial also improved, moving from an initial RMSE of 249.5 to 166.4, though it remained behind the polynomial kernel methods. The relative stability of all four optimised models is reflected in **Visualisation 09**, where resampled distributions are tight with few extreme values, indicating consistent behaviour across folds.

Training time for the optimised models is summarised in **Visualisation 10**. The expanded grid search came at a considerable computational cost for several methods. svmPoly required 81 minutes 29 seconds to train, and gaussprPoly and svmRadial required 14 minutes 53 seconds and 28 minutes 35 seconds respectively. brnn, despite searching across a larger neuron range than in the initial run, trained in 8 minutes 47 seconds, a substantially shorter time than the other three optimised models. This further reinforces brnn as the preferred method, delivering superior predictive performance with comparatively modest computational cost.

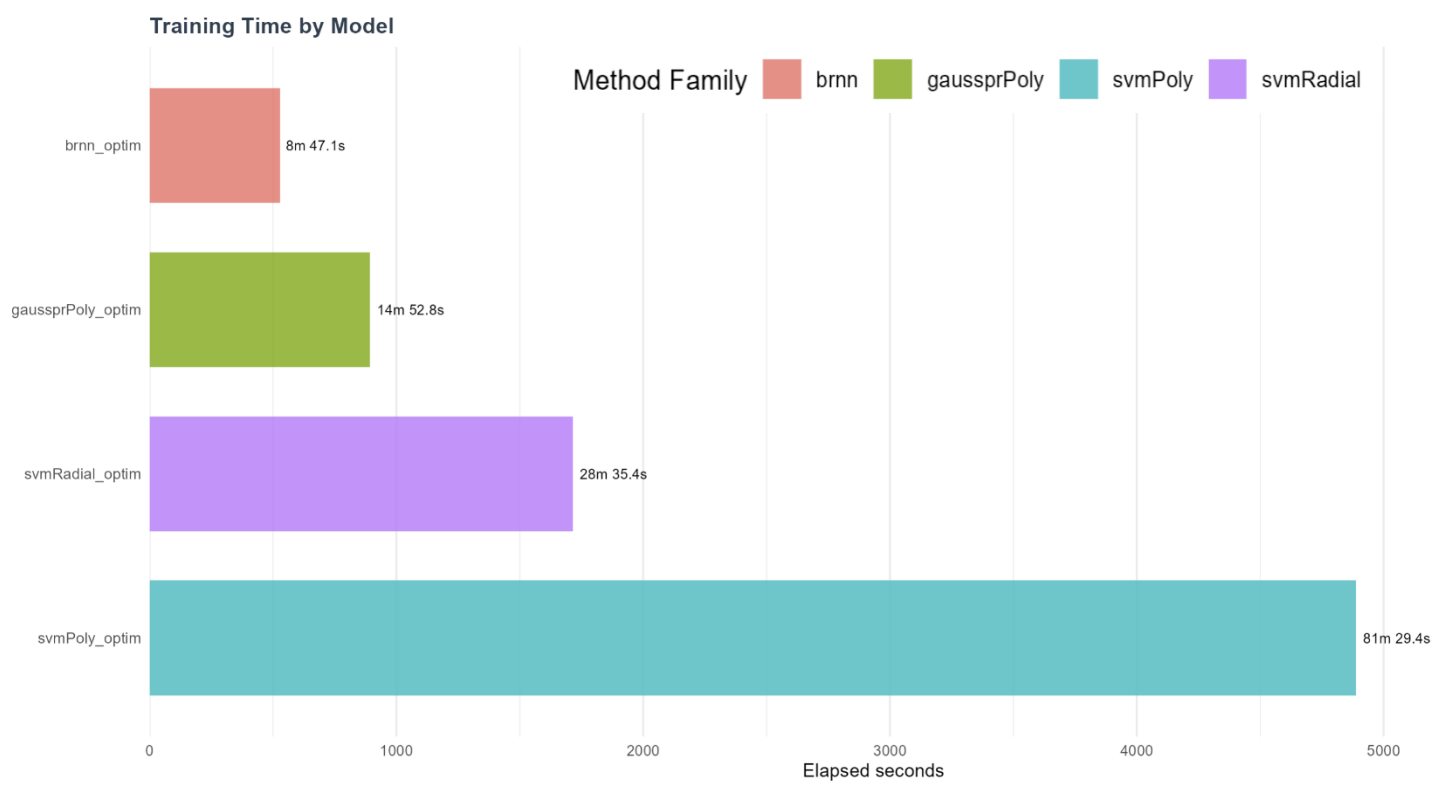
Table 04: Summary Metric Table of Optimised Methods								
Model	RMSE	RMSESD	R ²	R ² SD	MAE	MAESD	BestTune	Train Time
brnn_optim	110.5016	9.5108	0.9858	0.0026	82.2316	6.2551	neurons=5	8m 47.1s
svmPoly_optim	152.3937	12.9129	0.973	0.0048	110.3579	7.2095	degree=3, scale=0.01, C=20	81m 29.4s
gaussprPoly_optim	154.8437	13.3781	0.9725	0.005	109.6431	6.8778	degree=3, scale=0.05	14m 52.8s
svmRadial_optim	166.4274	15.681	0.9683	0.0054	115.484	6.4609	sigma=0.005, C=50	28m 35.4s
null	923.1446	32.9413	N/A	N/A	723.8446	26.2989	parameter=none	1.1s

Table 04: Summary results table of the optimised models. Cross-validated performance metrics for the four optimised models and the null baseline (highlighted in green), including RMSE, MAE, and R² alongside their standard deviations across resampled folds. All four models show improvement over their initial candidate run results, with brnn_optim remaining the strongest performer (highlighted in yellow).



Visualisation 09: Optimised model performance across resampled trials boxplots of cross-validated MAE, RMSE, and R² for the four optimised models and the null baseline, scaled relative to the null model. All optimised models display tight resampled distributions with few extreme values (that were relatively close), reflecting consistent and stable predictive behaviour across folds. brnn_optim throughout all three metrics outperformed the other three optimised models with an RSME of 110.50, MAE of 82.23, and R² of 0.99.

Training Time Comparison



Visualisation 10: Bar chart of training time (in elapsed minutes) for each optimised method. The brnn_optim method required only 9 minutes to train, due to fewer hyperparameters needing tuning. Training took approximately 15 minutes for gaussprPoly_optim and around 29 minutes for svmRadial_optim. The svmPoly_optim method required substantially more computation time (over 81 minutes) because of its expanded grid searches and greater number of hyperparameters.

Part D: Best Model

1). Optimised Models' Performance on Unseen Data:

Table 05 presents the training, bootstrap, and testing performance metrics for the four optimised models. Comparing performance across all three evaluation stages highlights clear differences in how effectively each model generalised to unseen data.

Overall, `brnn_optim` demonstrated the strongest and most stable generalisation performance. The model achieved a training RMSE of 73.8, which increased only modestly to 79.7 on the testing set, representing an 8.03% increase. Similarly, the testing R^2 remained very high at 0.991 compared with 0.994 during training. The bootstrap RMSE estimate of 110.5 (± 9.5) further supports the model's stability, with the relatively small standard deviation indicating consistent performance across resampled datasets. Collectively, these results suggest that `brnn_optim` generalised well with minimal evidence of overfitting.

In contrast, `gaussprPoly_optim` achieved the strongest training performance, producing a training RMSE of 62.5 and an R^2 of 0.996, indicating an excellent fit to the training data. However, this improvement did not translate to unseen observations. Testing RMSE increased substantially to 106.8, representing a 71.05% increase relative to training performance, while testing R^2 declined to 0.985. The bootstrap RMSE of 154.8 (± 13.4) also reflects weaker generalisation performance. Although the testing results remain reasonably strong overall, the large gap between training and testing performance suggests a noticeable degree of overfitting compared with `brnn_optim`.

A similar trend was observed for `svmPoly_optim`. Training RMSE increased from 74.0 to 116.4 on the testing set, corresponding to a 57.18% increase, while R^2 declined from 0.994 to 0.982. The bootstrap RMSE of 152.4 (± 12.9) is consistent with this reduction in generalisation performance. Although `svmPoly_optim` performed competitively during training, this performance advantage did not persist on unseen data. Together with `gaussprPoly_optim`, the substantial increase in testing error indicates clear overfitting, particularly as testing RMSE exceeded training RMSE by more than 15%.

`svmRadial_optim` produced the weakest training performance among the four models, with a training RMSE of 84.6. Despite this, the model still showed a considerable generalisation gap, with testing RMSE increasing to 127.4, representing a 50.62% increase, and testing R^2 declining from 0.992 to 0.979. The bootstrap RMSE of 166.4 (± 15.7) was also the highest observed, while the larger standard deviation suggests slightly less stability across resampled folds relative to the other models.

Overall, although `gaussprPoly_optim` and `svmPoly_optim` achieved lower training errors, these improvements were not maintained on the testing data. By comparison, `brnn_optim` showed the closest agreement between training, bootstrap, and testing performance, indicating superior generalisation and the strongest predictive stability across unseen observations. On this basis, `brnn_optim` was selected as the preferred model for final evaluation.

Table 05: Optimised Model Metrics										
Model	Training RMSE	Bootstrap RMSE ($\pm$ SD)	Testing RMSE	RMSE Difference	RMSE Change (%)	Training MAE	Testing MAE	Training R^2	Testing R^2	R^2 Difference
brnn_optim	73.7942	110.5016 ($\pm$ 9.5108)	79.7221	5.9279	8.03%	56.5085	63.3377	0.9937	0.9916	-0.002
gaussprPoly_optim	62.4576	154.8437 ($\pm$ 13.3781)	106.8367	44.3791	71.05%	45.7128	79.8436	0.9955	0.985	-0.0105
svmPoly_optim	74.034	152.3937 ($\pm$ 12.9129)	116.369	42.3354	57.18%	64.0517	84.6607	0.9936	0.9822	-0.0114
svmRadial_optim	84.5964	166.4274 ($\pm$ 15.6810)	127.4224	42.826	50.62%	68.6589	86.3123	0.9917	0.9786	-0.013

Table 05: Table 05: Predictive performance comparison of the four optimised regression models across the training, bootstrap resampling, and testing datasets. Model performance was evaluated using RMSE, MAE, and R^2 , where lower RMSE and MAE values indicate lower prediction error, and higher R^2 values indicate stronger explanatory power. Overall, brnn_optim demonstrated the strongest balance between predictive accuracy and generalisation performance, producing the lowest testing RMSE (79.72) while maintaining a very high testing R^2 value (0.9916). The relatively small difference between training and testing performance further suggests minimal overfitting and strong generalisation to unseen data. In comparison, the remaining models showed substantially larger increases in testing error, indicating weaker generalisation performance despite competitive training results.

2). Observations that do not Fit the Optimised Models:

Residual analysis further reinforced the superior performance of `brnn_optim`. Compared with the alternative models, `brnn_optim` produced smaller and more controlled residuals overall, indicating more stable predictive behaviour across patients. The largest positive residual for `brnn_optim` occurred for patient *tid57808* (Residual = 300.39), while the largest negative residual occurred for patient *tid57422* (Residual = -271.40). Despite these larger errors, the residual spread remained comparatively balanced, suggesting reduced systematic prediction bias.

In contrast, the remaining models produced substantially larger residual errors, with several patients repeatedly appearing as poorly predicted observations across multiple methods. Patient *tid57689* was the clearest example of this pattern, producing the largest positive residual in `gaussprPoly_optim` (510.14), `svmPoly_optim` (649.36), and `svmRadial_optim` (970.58). Notably, this patient did not appear among the major residual outliers for `brnn_optim`, suggesting that the BRNN model handled this observation considerably more effectively than the alternative approaches.

A similar pattern was observed for patient *tid57808*, which appeared as a large positive residual across all models, including `gaussprPoly_optim` (490.60), `svmPoly_optim` (454.03), `svmRadial_optim` (427.33), and `brnn_optim` (300.39). However, `brnn_optim` still produced the smallest residual for this patient, indicating comparatively better predictive accuracy even for difficult observations.

Additional recurring outliers included *tid57450* and *tid57695*, both of which appeared across several non-BRNN models. In particular, *tid57695* produced large negative residuals for `gaussprPoly_optim` (-275.63), `svmPoly_optim` (-393.50), and `svmRadial_optim` (-237.36), further suggesting that certain observations consistently challenged the kernel-based and Gaussian process approaches.

Overall, the repeated appearance of the same patients across the non-BRNN models suggests that particular observations contained characteristics that were difficult for these methods to model accurately. By comparison, the tighter and more stable residual distribution observed for `brnn_optim` indicates that its predictions generally remained closer to the true observed values. This aligns closely with the lower testing RMSE and stronger generalisation performance reported in **Table 05**, further supporting `brnn_optim` as the best-performing optimised model.

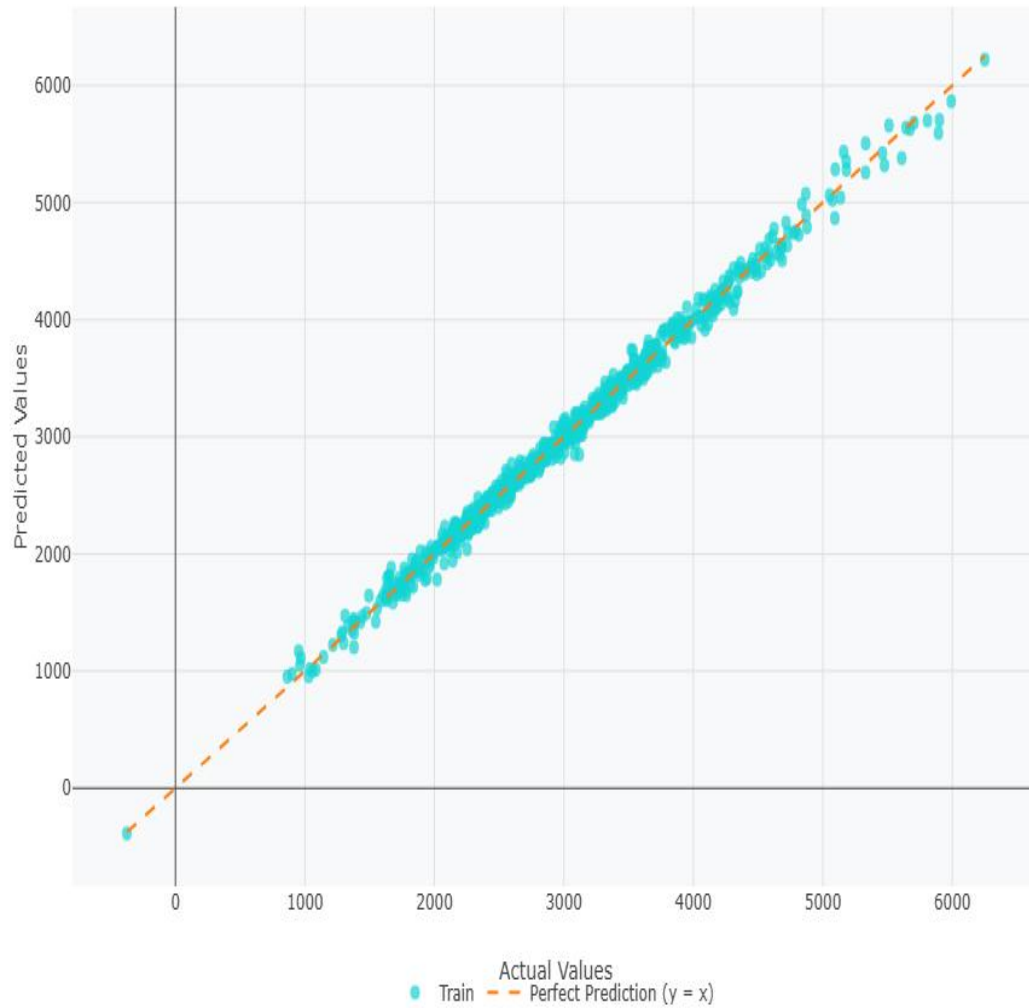
3). Further Best Model Assessment:

Visualisations 11 and 12 present the predicted-versus-actual plots for the training and testing datasets of the final brnn_optim model. In both datasets, predicted values aligned closely with the observed values, with most observations clustering tightly around the 45-degree reference line. This indicates strong predictive accuracy and demonstrates that the model was able to capture the underlying relationship within the data effectively. Importantly, the testing dataset showed a pattern highly consistent with the training dataset, suggesting strong generalisation performance with minimal evidence of overfitting.

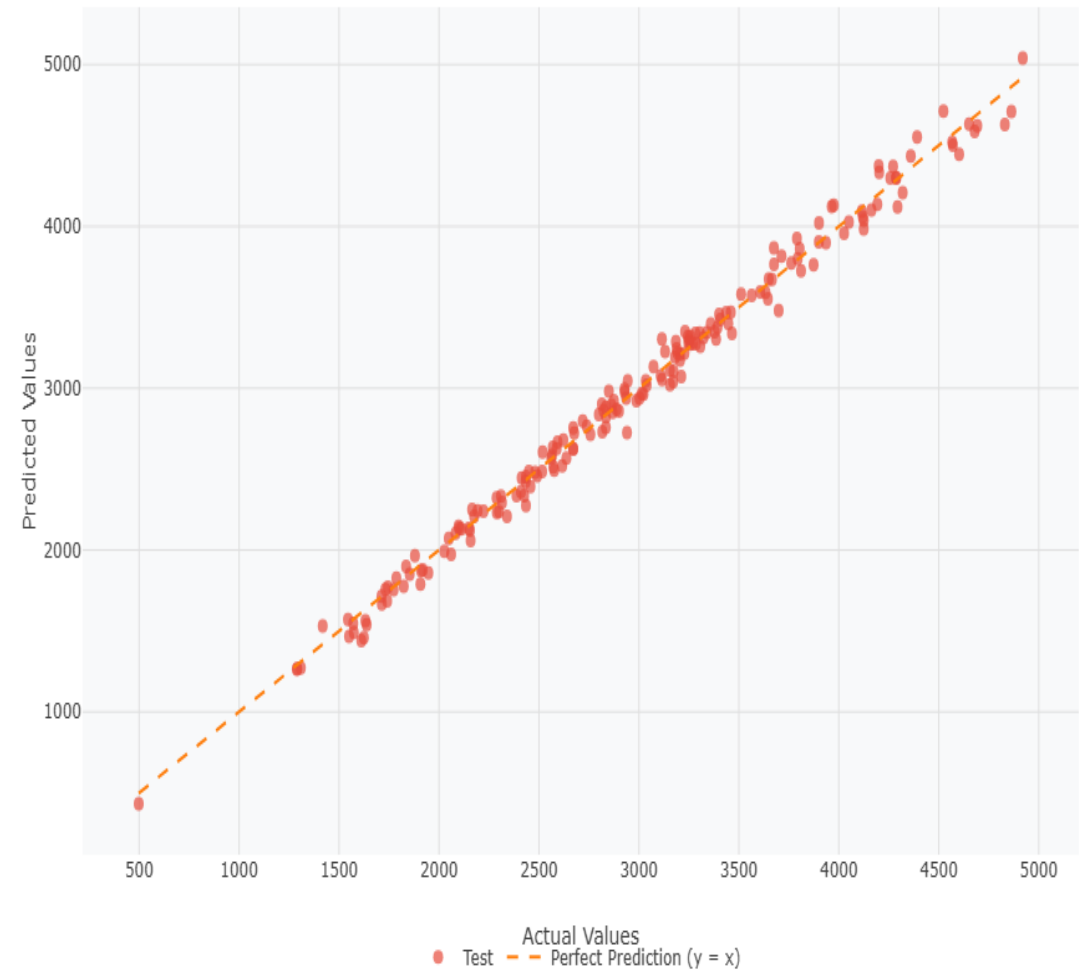
Further insight into model performance is provided through the residual analyses shown in **Visualisations 13 to 15**. **Visualisation 13 and 14** illustrates the residual scatterplot for both datasets, where residuals remained largely centred around zero with relatively limited dispersion, indicating balanced prediction errors and stable predictive behaviour across observations. Similarly, the residual boxplots shown in **Visualisation 15** demonstrated relatively compact residual distributions with only a limited number of observations extending beyond the $1.5 \times \text{IQR}$ threshold. Collectively, these findings suggest that most prediction errors were comparatively small, contributing to the model's low overall RMSE and strong predictive performance.

The relatively small number of residual outliers identified through the $1.5 \times \text{IQR}$ criterion is summarised in **Table 06**. Although several observations produced larger residuals, these cases were infrequent relative to the full dataset and therefore had limited impact on overall model performance. Together, **Visualisations 11–15** and **Table 06** provide strong evidence that the brnn_optim model achieved accurate and stable predictions across both the training and testing datasets.

Predicted vs Actual Values



Predicted vs Actual Values

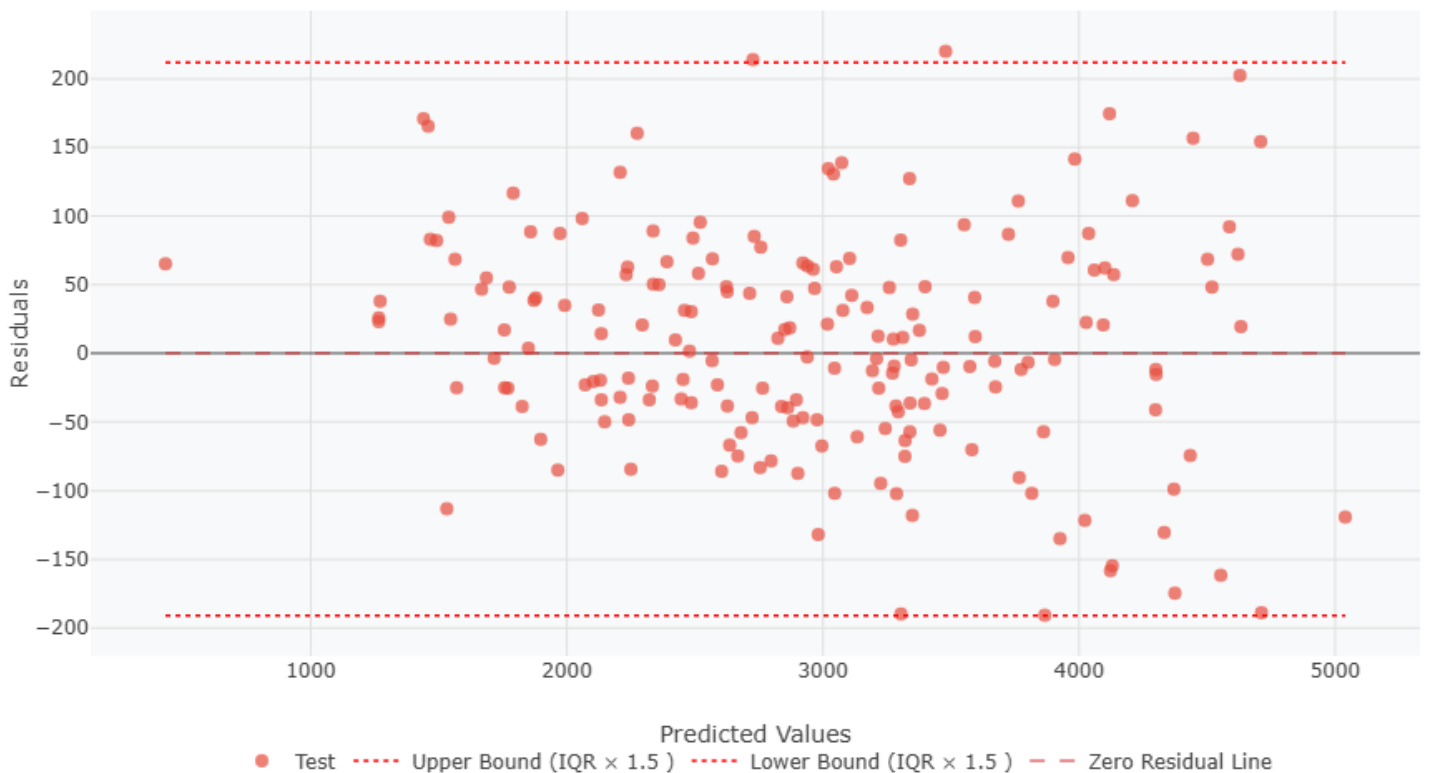


Visualisations 11 (left showing train) and 12 (right showing test): Predicted-versus-actual plots for the training and testing datasets of the final brnn_optim model. In both datasets, predicted values aligned closely with the observed values, with most observations clustering near the 45-degree reference line. This indicates strong predictive accuracy and consistent model performance across both seen and unseen data. The similarity between the training and testing plots further suggests that the model generalised effectively with minimal evidence of overfitting.

Residual Analysis - IQR Multiplier: 1.5

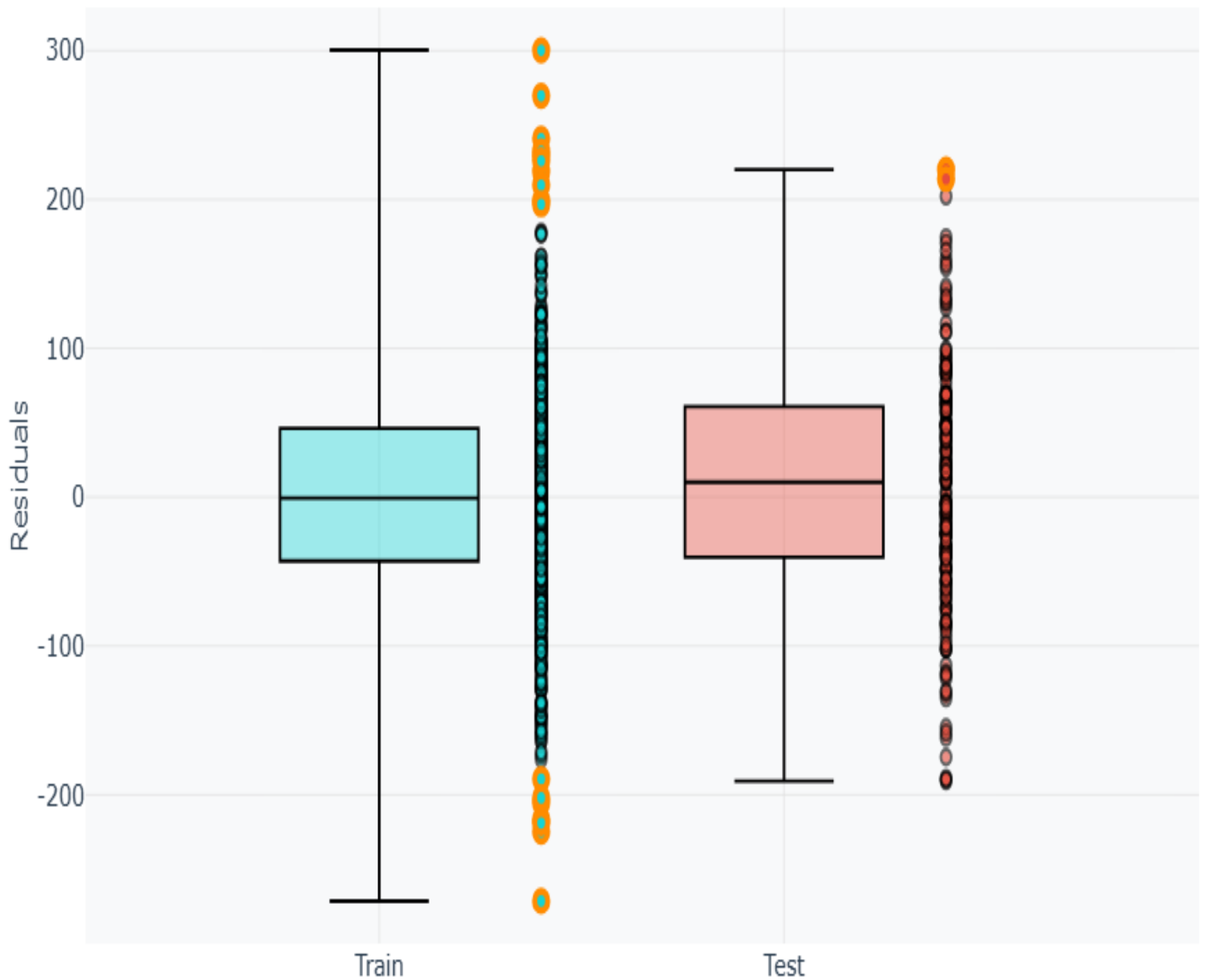


Residual Analysis - IQR Multiplier: 1.5



Visualisation 13 (top showing train) and 14 (bottom showing test): Residual scatterplot for the training and testing datasets of the final brnn_optim model. Residuals were generally centred around zero with relatively limited dispersion, indicating balanced prediction errors and stable predictive behaviour across observations. Only a relatively small number of observations exceeded the $1.5 \times IQR$ residual threshold, suggesting that large prediction errors were uncommon overall. These identified residual outliers are summarised in **Table 06**.

Residual Boxplot - IQR Multiplier: 1.5



Visualisation 15: Boxplot of residual distributions for the training and testing datasets of the final brnn_optim model. The residual distributions remained relatively compact and centred near zero, indicating limited overall prediction error and minimal systematic bias. While several outlying residuals were detected beyond the $1.5 \times \text{IQR}$ threshold, the number of extreme observations (as highlighted in orange) remained comparatively small relative to the full dataset. The specific residual outliers identified through this analysis are reported in **Table 06**.

Table 06: Residual Outliers for brnn_optim (IQR = 1.5)

Patient	Dataset	Actual	Predicted	Residual	Direction
tid-57808	Train	5,892.58	5,592.19	300.3931	Above
tid-57810	Train	3,119.15	2,849.39	269.7661	Above
tid-57378	Train	2,020.97	1,780.02	240.9433	Above
tid-57967	Train	3,083.68	2,851.37	232.3044	Above
tid-57154	Train	5,608.72	5,379.38	229.3455	Above
tid-58045	Train	5,092.44	4,866.34	226.102	Above
tid-57118	Test	3,698.96	3,478.89	220.0615	Above
tid-57486	Train	4,310.47	4,091.66	218.814	Above
tid-57825	Test	2,940.37	2,726.51	213.8604	Above
tid-57920	Train	2,252.21	2,042.44	209.7721	Above
tid-57905	Train	2,141.96	1,942.47	199.492	Above
tid-57921	Train	5,900.13	5,703.22	196.9115	Above
tid-57832	Train	5,095.50	5,284.88	-189.3714	Below
tid-57947	Train	3,534.15	3,736.19	-202.0354	Below
tid-57682	Train	4,868.05	5,073.35	-205.3025	Below
tid-57668	Train	1,665.84	1,882.62	-216.7763	Below
tid-57930	Train	951.4609	1,170.43	-218.9708	Below
tid-57894	Train	3,520.08	3,744.90	-224.8188	Below
tid-57422	Train	5,160.25	5,431.64	-271.3963	Below

Table 06: Observations identified as residual outliers for the final brnn_optim model using the $1.5 \times \text{IQR}$ criterion applied to the residual distributions shown in **Visualisations 13 to 15**. The table reports the patient identifier, dataset membership, actual value, predicted value, residual magnitude, and residual direction. Although several observations exhibited comparatively large residuals, the number of detected outliers remained relatively small compared with the overall dataset, supporting the model's strong predictive stability and low overall prediction error.

4). Best Model Theory Explanation:

The BRNN was implemented using the `brnn` package in R (Perez-Rodriguez and Gianola, 2025), which provides an efficient implementation of Bayesian regularised feedforward neural networks within the *caret* framework. The model is based on a standard feedforward artificial neural network architecture, which is capable of modelling complex and non-linear relationships between predictor variables and an outcome. Neural networks achieve this by passing inputs through interconnected layers of processing units (neurons), allowing the model to learn flexible functional forms that do not need to be explicitly specified in advance.

In a standard feedforward neural network, predictions are generated through weighted combinations of predictor variables passed through non-linear activation functions. A single hidden-layer neural network can be expressed mathematically as:

$$\hat{y} = \beta_0 + \sum_{j=1}^m w_j \phi \left(\alpha_{j0} + \sum_{i=1}^p \alpha_{ji} x_i \right)$$

where x_i represents the predictor variables, α_{ji} are the weights connecting inputs to hidden neurons, w_j are the weights connecting hidden neurons to the output layer, $\phi(\cdot)$ is a non-linear activation function, m is the number of hidden neurons, and $\hat{y}$ is the predicted response value. Unlike traditional linear regression models, this structure allows the network to automatically learn non-linear patterns and interaction effects directly from the data without requiring explicit specification of transformation or interaction terms.

Building on this framework, the Bayesian Regularised Neural Network (BRNN) extends the standard neural network by introducing a probabilistic regularisation approach to prevent overfitting and improve generalisation. Rather than minimising prediction error alone, the model minimises a total objective function denoted by F , which represents the overall quantity being optimised during training:

$$F = \beta E_D + \alpha E_W$$

where E_D is the sum of squares error measuring the difference between predicted and observed values, and E_W is the sum of squared network weights and biases, representing model complexity. The parameter β controls the influence of the data error term, determining how strongly the model prioritises fitting the observed training data, while α controls the strength of the penalty applied to large weights, encouraging simpler model structures. Together, these terms define a trade-off between data fit and model complexity.

From a probabilistic perspective, this formulation can be interpreted as placing Gaussian prior distributions over the network weights. In this context, $\beta = \frac{1}{2\sigma_e^2}$ is associated with the inverse of the noise variance in the observed data, while $\alpha = \frac{1}{2\sigma_\theta^2}$ relates to the inverse of the prior variance of the network weights. This links the optimisation process directly to assumptions about data noise and expected parameter variability. Rather than being manually specified, both parameters are estimated during training using Bayesian evidence procedures, which also compute the effective number of parameters γ , representing how many weights meaningfully contribute to explaining signal in the data rather than noise (MacKay, 1992, p. 460).

In the implementation used within the `brnn` package, training is performed using a Gauss–Newton approximation to backpropagation, which avoids the computational cost of calculating the full Hessian matrix while still enabling efficient updates of the Bayesian regularisation parameters (Foresee &

Hagan, 1997). This makes the method computationally practical while preserving the theoretical advantages of Bayesian learning.

Turning towards model performance, the relatively small gap between training RMSE (73.8) and testing RMSE (79.7) suggests the regularisation was functioning as intended, with the model generalising well rather than overfitting the training sample. By comparison, models such as the polynomial Gaussian Process regression (`gaussprPoly_optim`) showed substantially larger increases in error on unseen data (for example in `gaussprPoly_optim`'s case, it showed a 71.05% increase in RMSE on the testing data), indicating overfitting. The BRNN avoids this because it automatically adjusts its effective complexity based on what the data supports, rather than relying on fixed kernel or tuning parameters. The bootstrap standard deviation of ± 9.5 further supports this interpretation, indicating stable performance across resampled training sets.

The characteristics of the dataset also appear well suited to this modelling approach, with nonlinear but relatively smooth relationships between predictors and the response variable. This structure aligns well with a regularised neural network, which is flexible enough to capture nonlinear patterns while remaining constrained by Bayesian regularisation. Residual diagnostics further support this, with errors generally centred around zero and few extreme values, suggesting the model was not overly influenced by noise or outliers.

Part E: Transparent Model

1). Transparent Model Development:

Although the optimised `brnn_optim` achieved the strongest predictive performance overall, its neural network structure provides limited transparency regarding the exact calculations and relationships contributing to individual predictions. Given the medical context of this dataset, model interpretability is an important consideration, as there should be some capacity to explain how particular predictions are produced and which predictor relationships contribute to the outcome.

Inspection of the optimised modelling approaches indicated that several high-performing methods, including `svmPoly_optim` and `gaussprPoly_optim`, made effective use of higher-order relationships through the inclusion of polynomial terms, both favouring a hyperparameter setting of degree = 3. Motivated by this observation, a transparent modelling approach was explored by extending a `glmnet` framework to include interaction terms, allowing the model to capture more complex predictor relationships while retaining coefficient-based interpretability. By incorporating interaction effects, the model was also able to move beyond strict linear assumptions and better represent situations where the effect of one predictor variable depended on the value of another predictor.

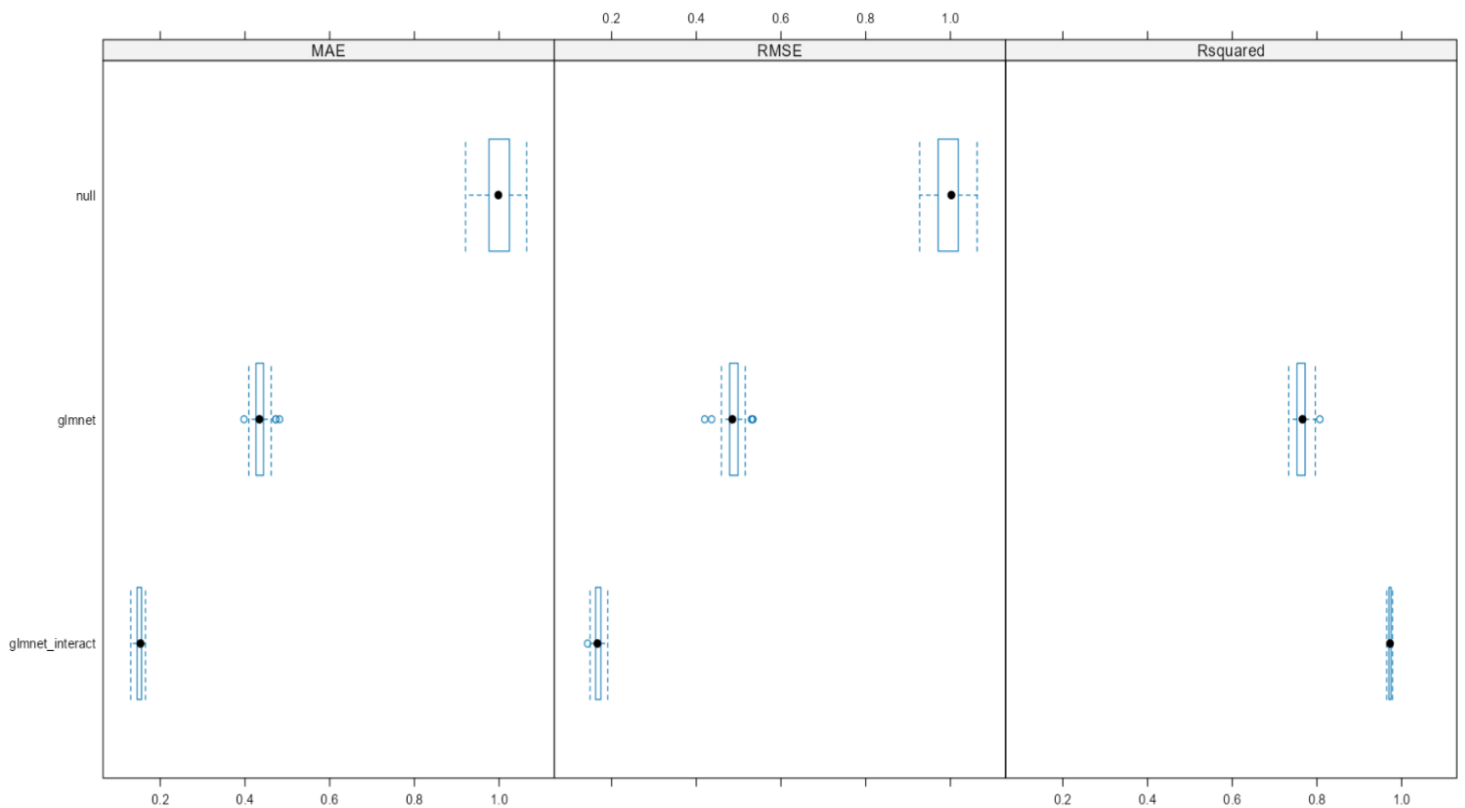
The resulting bootstrap training performance is summarised in **Table 07**. The standard `glmnet` model without interaction terms produced moderate predictive performance, with a bootstrap RMSE of 449.31 (± 24.05), $R^2 = 0.76$, and MAE of 316.61. However, incorporating interaction terms resulted in a substantial improvement across all performance metrics. The `glmnet_interact` model achieved a bootstrap RMSE of 154.24 (± 9.61), $R^2 = 0.97$, and MAE of 108.77, considerably outperforming the non-interaction version while also demonstrating substantially greater stability, as evidenced by the lower RMSE standard deviation. This improvement relative to the null model is visually evident within **Visualisation 16**, where `glmnet_interact` consistently achieved substantially lower error metrics and stronger explanatory power than the standard `glmnet` approach.

Although the interaction-based approach required a substantially longer computational training time, the predictive improvement was considerable. As shown in **Visualisation 17**, the standard `glmnet` model required approximately 40 seconds to train, whereas the `glmnet_interact` model required nearly 9 minutes due to the significantly increased number of generated interaction features. Despite this increase in computational complexity, the resulting performance gain justified the additional training cost.

Table 07: Summary Metric Table of Transparent Model

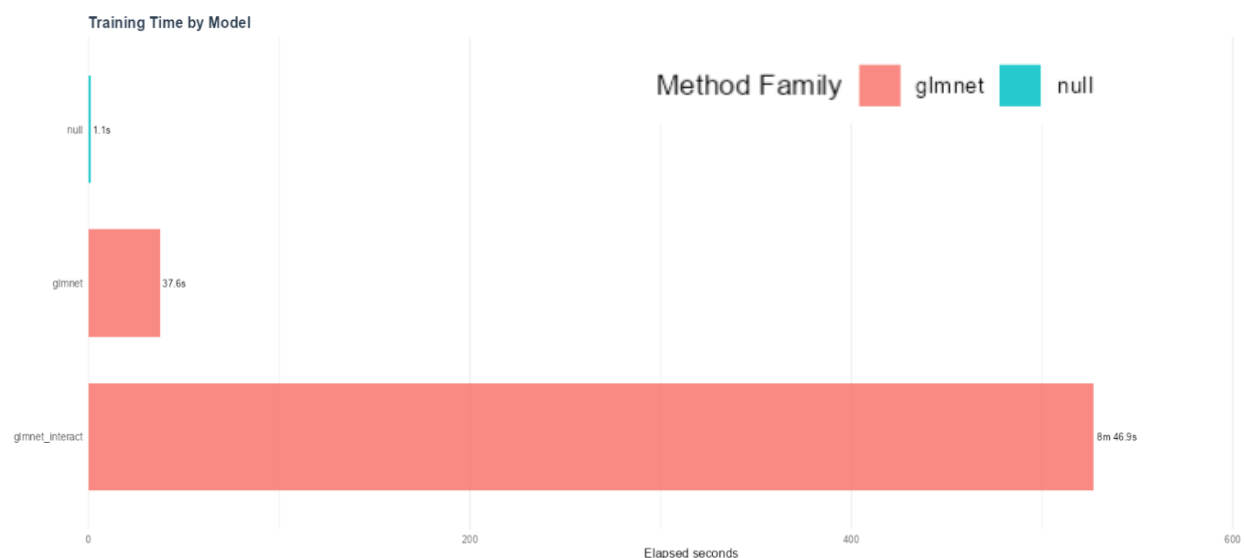
Method	RMSE	RMSESD	R ²	R ² SD	MAE	MAESD	BestTune	Train Time
GLMNET (no interact)	449.31	24.05	0.76	0.02	316.61	14.75	alpha = 0.55, lambda = 10.44	37.6s
GLMNET (with interact)	154.24	9.61	0.97	0	108.77	5.88	alpha = 1, lambda = 0.73	8min 47s
Null model	923.14	32.94	NA	NA	723.84	26.3	—	1.1s

Table 07: Demonstration of the substantial improvement achieved after incorporating interaction terms into the glmnet framework. The glmnet_interact model produced considerably lower RMSE and MAE values alongside substantially higher R², indicating stronger predictive performance and improved explanatory capability compared with the standard glmnet model. Additionally, the lower RMSE standard deviation suggests that the interaction model produced more stable performance across bootstrap resamples.



Visualisation 16: Comparison of RMSE, MAE, and R^2 performance relative to the null model across both transparent modelling approaches. The glmnet_interact model demonstrates clear performance improvements over the standard glmnet model, achieving substantially lower prediction errors and stronger explanatory power. In addition, the reduced variability across metrics indicates greater model stability and consistency.

Training Time Comparison



Visualisation 17: Comparison of the computational training time required for each transparent modelling approach. While the standard glmnet model required approximately 40 seconds to train, the glmnet_interact model required nearly 9 minutes due to the large number of generated interaction terms. This highlights the increased computational complexity associated with modelling higher-order predictor relationships.

2). Transparent Models' Performance to Unseen Data:

When evaluated on unseen testing data, the transparent interaction model maintained strong generalisation performance as seen on **Table 08**. The glmnet model without interaction terms produced a testing RMSE of 458.22, a testing MSE of 331.94, and testing $R^2 = 0.724$, representing comparatively limited predictive capability. In contrast, the glmnet_interact model achieved a testing RMSE of 133.41, testing MAE of 89.51, and testing $R^2 = 0.977$. Although the interaction model demonstrated a larger percentage increase between training and testing RMSE (20.5%) compared with the simpler glmnet model (5.24%), the absolute RMSE difference was nearly identical between models (approximately 22.7 points).

The strong predictive capability of the glmnet_interact model is further demonstrated within **Visualisations 18 and 19**, where predicted values closely align with the 45-degree reference line across both the training and testing datasets. While several visually distinct residual outliers were observed, these occurred in both the positive and negative directions, indicating instances of both overprediction and underprediction. The largest negative residuals were observed for patients *tid-57777* (-547.71, Test), *tid-57147* (-525.56, Test), and *tid-57666* (-487.02, Train), whereas the largest positive residuals were observed for *tid-57537* (+611.33, Train), *tid-57306* (+604.11, Test), and *tid-57689* (+592.89, Test). Overall, the residual magnitudes remained relatively balanced, suggesting no strong systematic prediction bias.

While the transparent glmnet_interact model did not surpass the predictive performance of the optimised brnn_optim model, its results remain highly competitive. Importantly, it provides the added advantage of interpretability through an explicit coefficient-based structure and observable interaction effects, allowing clearer examination of the mathematical relationships contributing to prediction outcomes. This trade-off between maximal predictive accuracy and explainability is particularly relevant in medical modelling applications. A full list of the coefficients that are present in the glmnet_interact model can be found in the **Appendix 01**.

Residual analysis using the $1.5 \times \text{IQR}$ criterion further supported the robustness of the transparent interaction model. As shown in **Visualisation 20 and 21**, residuals were generally concentrated around zero with both positive and negative residual outliers identified across the training and testing datasets. Although slightly lower-tail residual outliers were detected, the overall distribution and magnitude of positive and negative residuals remained broadly comparable, suggesting that the model maintained relatively balanced predictive behaviour across most observations.

Similarly, the boxplot residual analysis shown in **Visualisation 22** identified several prominent residual outliers within both the upper and lower tails, particularly among testing observations highlighted in orange. These outliers correspond to patients where the model either substantially overpredicted or underpredicted the true response value. Despite these observations, the overall magnitude and frequency of extreme residuals remained substantially lower and more stable than those observed across several competing modelling approaches, supporting the suitability of the transparent interaction model as an interpretable alternative with strong predictive performance.

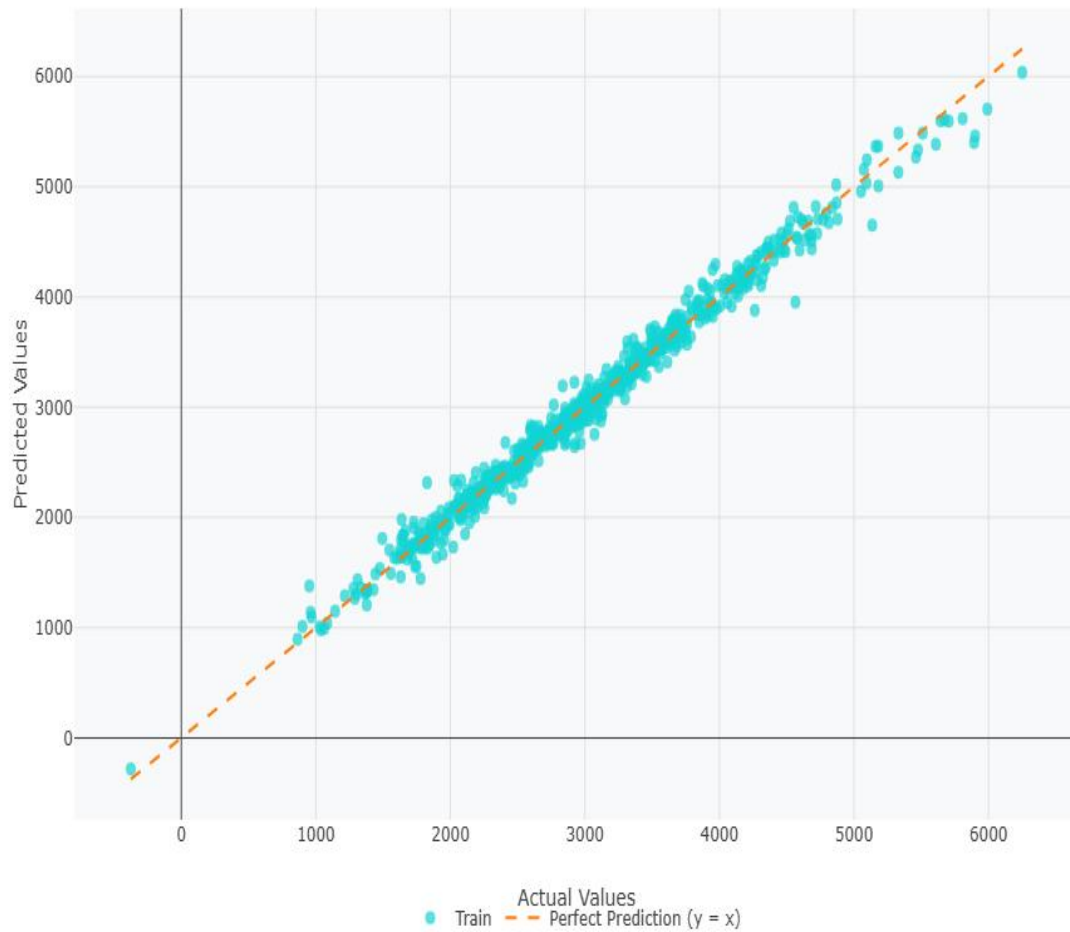
A full list of the residual outliers for both the train and test sets that were detected by **Visualisation 20 to 22** can be found using **Table 09**.

Table 08: Transparent Model Metrics

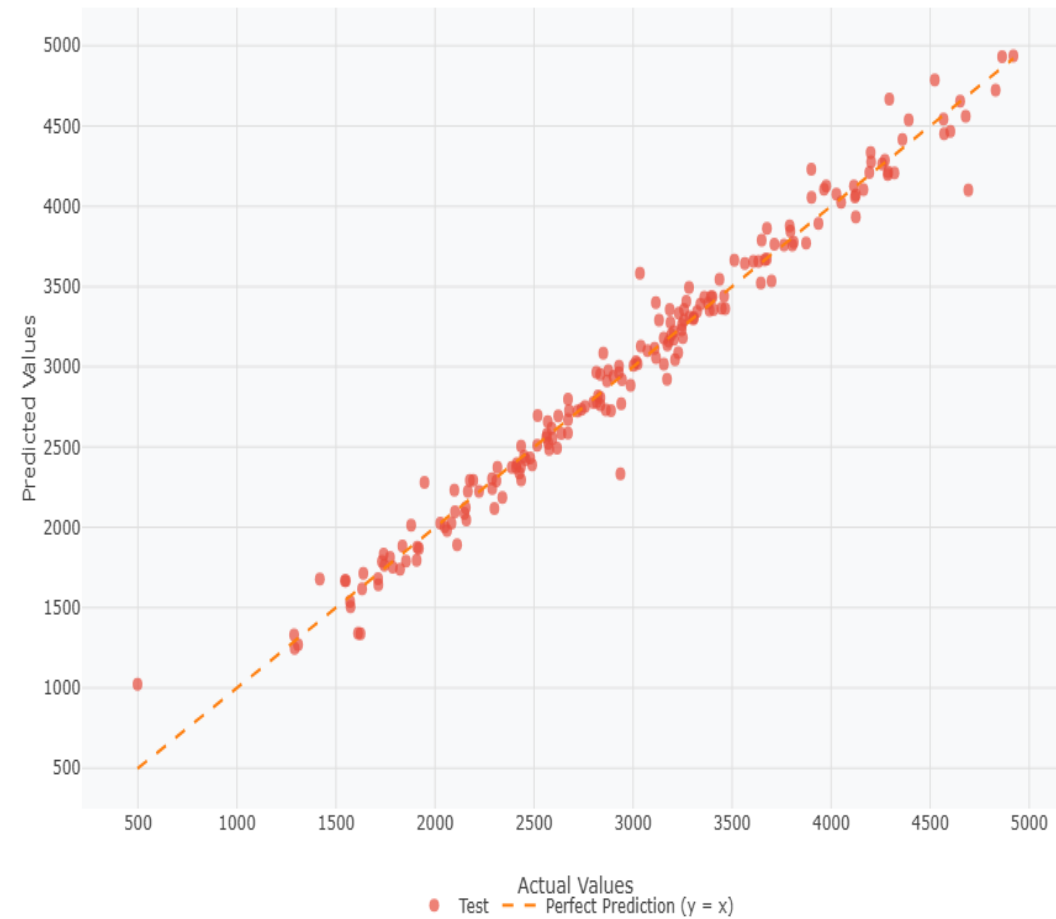
Model	Training RMSE	Bootstrap RMSE ($\pm$ SD)	Testing RMSE	RMSE Difference	RMSE Change (%)	Training MAE	Testing MAE	Training R ²	Testing R ²	R ² Difference
glmnet (no interact term)	435.4229	449.3128 ($\pm$ 24.0479)	458.2231	22.8001	5.24%	306.084	331.9378	0.7792	0.7236	-0.0557
glmnet (with interact term)	110.711	154.5116 ($\pm$ 9.6490)	133.4122	22.7011	20.50%	79.9924	89.5091	0.9857	0.9766	-0.0092

Table 08: Summary of the training and testing performance of the transparent modelling approaches. The glmnet_interact model substantially outperformed the standard glmnet model across all major evaluation metrics, achieving considerably lower RMSE and MAE values alongside substantially higher R². Although the interaction model showed a larger proportional increase in RMSE between training and testing datasets, overall predictive performance remained highly stable and generalised well to unseen observations.

Predicted vs Actual Values

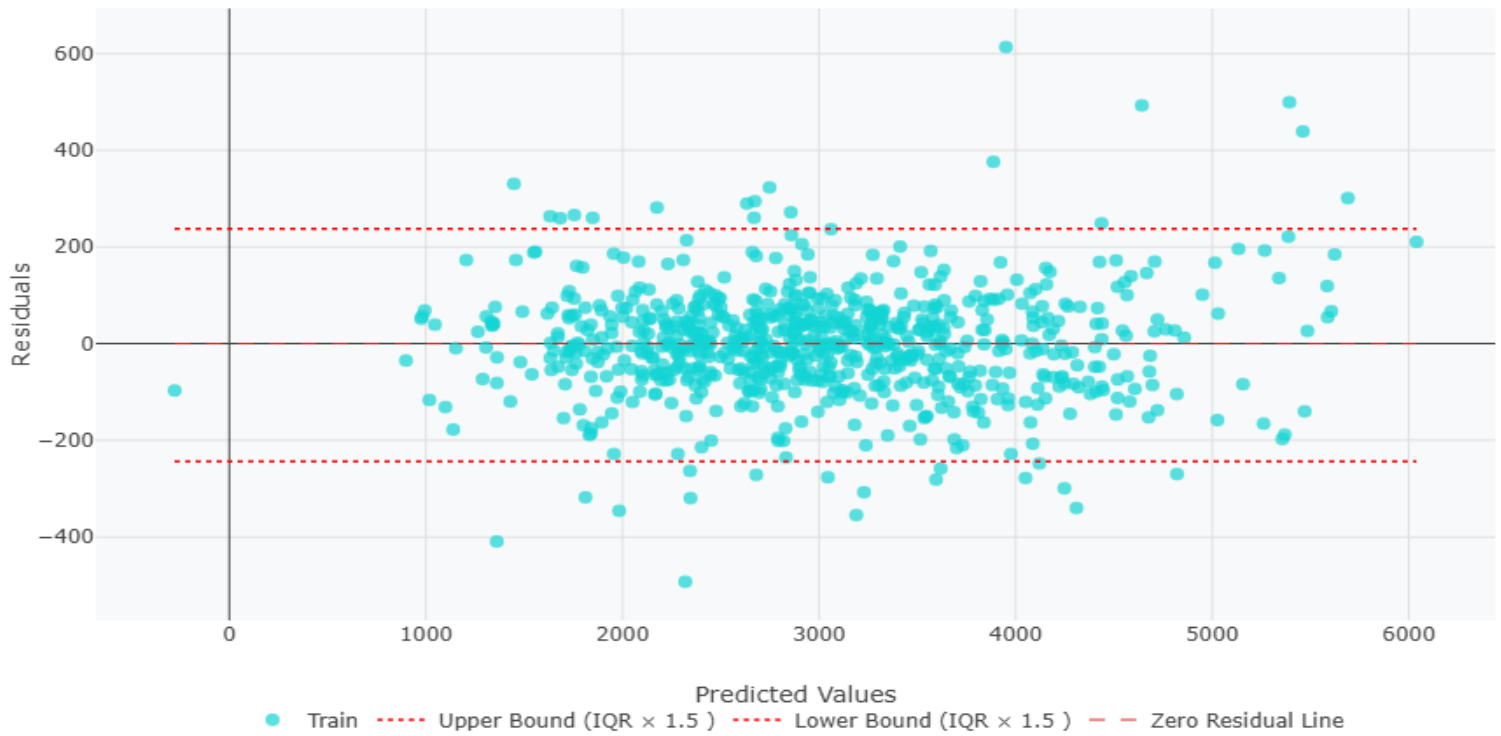


Predicted vs Actual Values

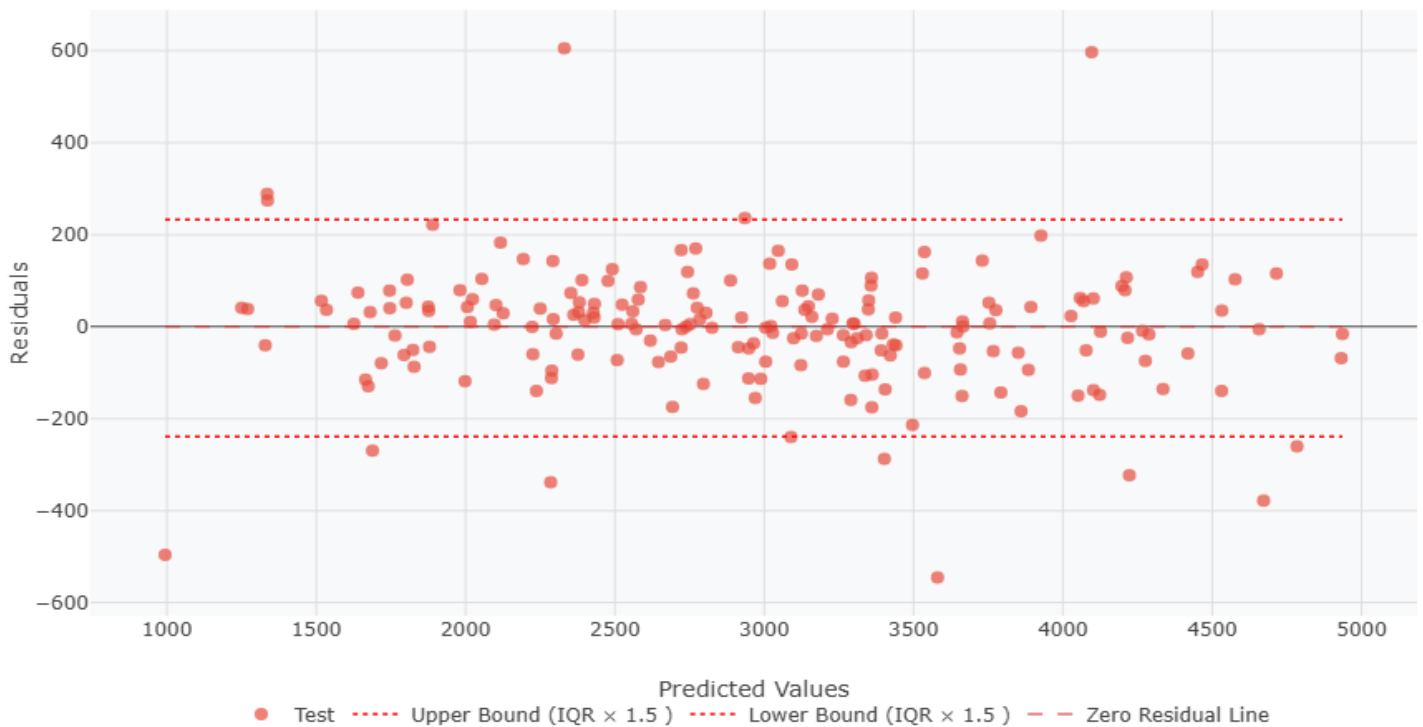


Visualisation 18 (left showing train) and 19 (right showing test): Comparison of the predicted and actual response values for the `glmnet_interact` model across the training and testing datasets. Most observations closely follow the 45-degree reference line, indicating strong predictive accuracy and good model calibration. Although several residual outliers are visible above and below the reference line, the majority of predictions remain tightly clustered around the line, supporting the strong predictive performance of the model.

Residual Analysis - IQR Multiplier: 1.5

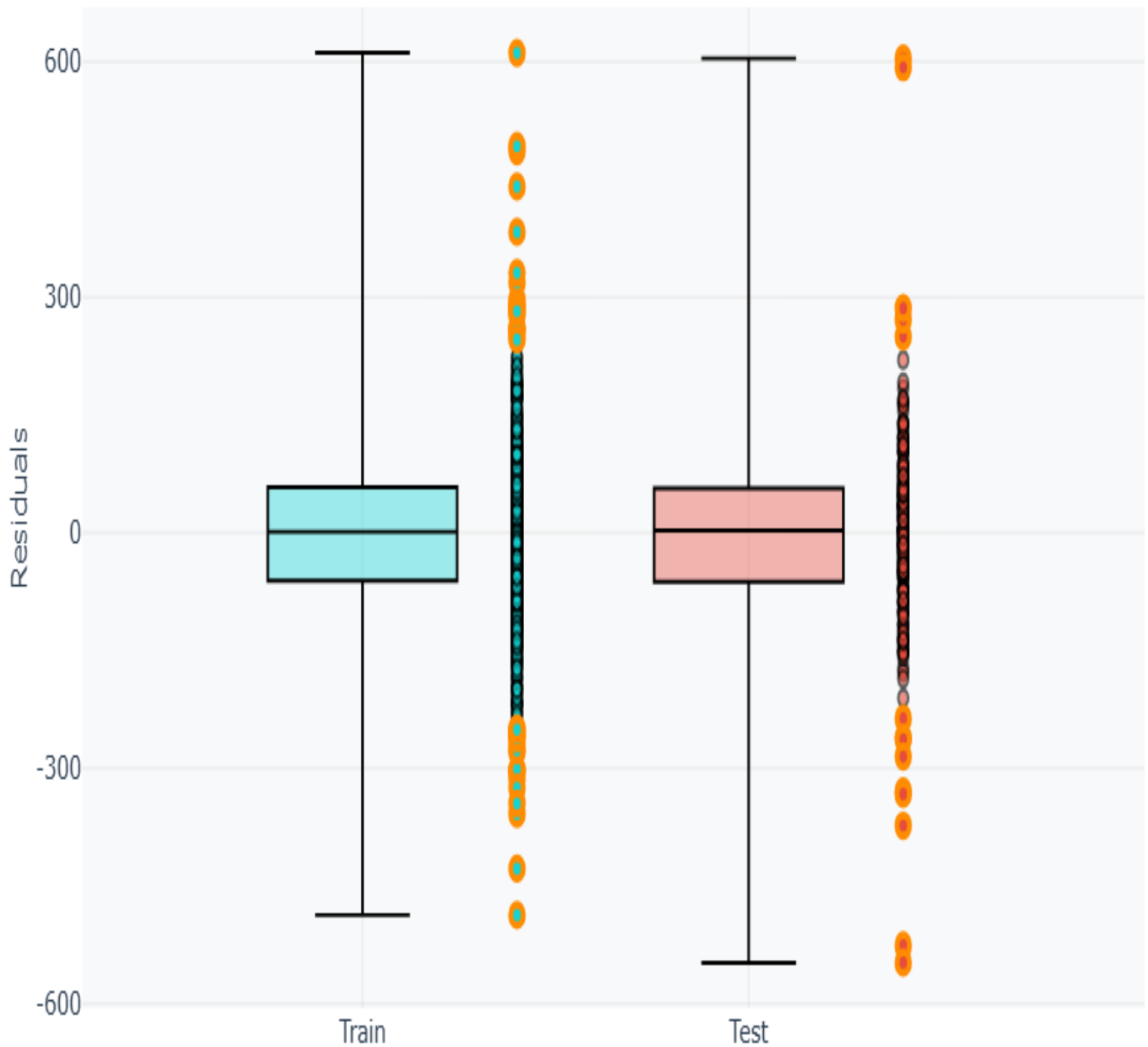


Residual Analysis - IQR Multiplier: 1.5



Visualisation 20 (top showing train) and 21 (bottom showing test): Presents the residual scatterplot analysis for the train and testing of the glmnet_interact model. Residuals are generally concentrated around zero, supporting strong overall model fit. Both positive and negative residual outliers are present, with the overall residual spread appearing relatively balanced across the upper and lower tails. This suggests that the model did not exhibit strong systematic overprediction or underprediction behaviour.

Residual Boxplot - IQR Multiplier: 1.5



Visualisation 22: Presents the residual distribution in boxplot form. Several prominent residual outliers are visible within both the upper and lower tails, particularly within the testing dataset. These observations indicate patients where the model produced comparatively large prediction errors. Despite these outliers, the overall residual spread remained comparatively constrained relative to several competing modelling approaches.

Table 09: Residual Outliers for glmnet_interact (IQR = 1.5)

Patient	Dataset	Actual	Predicted	Residual	Direction	Patient	Dataset	Actual	Predicted	Residual	Direction
tid-57537	Train	4,564.74	3,953.42	611.3259	Above	tid-57279	Test	2,849.61	3,086.06	-236.453	Below
tid-57306	Test	2,936.12	2,332.01	604.1094	Above	tid-57597	Train	3,872.98	4,122.38	-249.402	Below
tid-57689	Test	4,692.68	4,099.79	592.8865	Above	tid-57982	Train	2,768.62	3,019.20	-250.574	Below
tid-57808	Train	5,892.58	5,400.15	492.4303	Above	tid-57281	Train	3,360.59	3,617.01	-256.415	Below
tid-57747	Train	5,135.57	4,649.52	486.0498	Above	tid-57125	Train	2,080.20	2,338.89	-258.68	Below
tid-57921	Train	5,900.13	5,459.23	440.8951	Above	tid-57406	Test	1,418.16	1,679.24	-261.079	Below
tid-57449	Train	4,262.57	3,879.44	383.1334	Above	tid-57961	Train	4,551.20	4,813.29	-262.083	Below
tid-57720	Train	1,778.51	1,447.46	331.0573	Above	tid-57943	Test	4,523.26	4,786.20	-262.933	Below
tid-57178	Train	3,071.05	2,752.79	318.2581	Above	tid-57254	Train	2,408.98	2,678.21	-269.235	Below
tid-57119	Train	2,968.31	2,671.52	296.7949	Above	tid-57741	Train	3,314.43	3,591.73	-277.299	Below
tid-57378	Train	2,020.97	1,730.14	290.8271	Above	tid-57376	Train	3,772.44	4,051.24	-278.798	Below
tid-57282	Train	5,991.22	5,703.15	288.0672	Above	tid-57973	Test	3,114.96	3,399.99	-285.029	Below
tid-57799	Test	1,624.02	1,337.95	286.0705	Above	tid-57676	Train	3,949.14	4,250.01	-300.878	Below
tid-57845	Train	2,456.65	2,174.37	282.2816	Above	tid-57210	Train	2,921.74	3,224.76	-303.019	Below
tid-57520	Train	2,922.85	2,641.21	281.6318	Above	tid-57602	Train	2,026.77	2,332.74	-305.978	Below
tid-57192	Train	1,943.24	1,666.58	276.664	Above	tid-57490	Train	1,493.97	1,807.86	-313.887	Below
tid-57456	Test	1,611.32	1,339.87	271.45	Above	tid-57726	Train	3,969.70	4,294.38	-324.677	Below
tid-57157	Train	1,895.82	1,635.07	260.755	Above	tid-57455	Test	3,899.99	4,229.42	-329.433	Below
tid-57129	Train	2,109.33	1,849.11	260.2152	Above	tid-58022	Test	1,946.81	2,279.65	-332.839	Below
tid-57708	Train	2,929.86	2,670.91	258.9471	Above	tid-57787	Train	1,637.55	1,982.28	-344.735	Below
tid-57362	Train	4,685.77	4,435.19	250.582	Above	tid-57653	Train	2,834.82	3,193.01	-358.196	Below
tid-57450	Test	3,171.44	2,921.73	249.7067	Above	tid-57100	Test	4,293.62	4,666.65	-373.027	Below
tid-57810	Train	3,119.15	2,872.92	246.2296	Above	tid-57930	Train	951.4609	1,379.30	-427.839	Below
						tid-57666	Train	1,826.65	2,313.67	-487.022	Below
						tid-57147	Test	497.8689	1,023.42	-525.555	Below
						tid-57777	Test	3,034.47	3,582.18	-547.705	Below

Table 09: Summarises the major residual outliers identified for the glmnet_interact model using the $1.5 \times \text{IQR}$ criterion. Both overpredicted (“Above”) and underpredicted (“Below”) observations were detected, with residual magnitudes appearing relatively comparable across both directions. The largest positive residuals included patients *tid-57537* (+611.33), *tid-57306* (+604.11), and *tid-57689* (+592.89), while the largest negative residuals were observed for *tid-57777* (−547.71), *tid-57147* (−525.56), and *tid-57666* (−487.02).

Overall, the residual outliers indicate that although the glmnet_interact model achieved strong predictive performance across most observations, a small subset of patients remained difficult to model accurately. However, the residual distribution remained relatively balanced overall, with no strong evidence of systematic prediction bias. Despite the presence of several larger residuals, the magnitude and frequency of extreme residuals remained comparatively constrained relative to several alternative modelling approaches.

3). Transparent Model Theory Explanation:

A major advantage of the `glmnet_interact` model compared with the optimised `brnn_optim` model is that the influence of individual predictors can be directly examined through the model coefficients. While the `brnn` model achieved stronger predictive performance overall, its neural network architecture functions largely as a black-box model, meaning that the internal relationships contributing to predictions are difficult to interpret directly. Rudin (2019) argues that in high-stakes decision domains such as healthcare, this opacity is a fundamental problem rather than just an inconvenience, as the ability to examine and challenge a model's reasoning is a core requirement of responsible deployment. In contrast, the `glmnet_interact` model retains a transparent coefficient-based structure where the mathematical contribution of each predictor variable can be explicitly quantified and examined.

The `glmnet` algorithm belongs to the family of penalised regression models introduced by Zou and Hastie (2005) and implemented within the `glmnet` framework developed by Friedman et al. (2010). The method combines both L1 (lasso) and L2 (ridge) regularisation penalties within a single optimisation framework, allowing the model to simultaneously perform coefficient shrinkage and variable selection. The L1 component of this penalty is particularly important for interpretability. Tibshirani (1996) demonstrated that the L1 penalty has the mathematical property of shrinking some coefficients to exactly zero, allowing for the removal of irrelevant predictors from the model. Predictors that contribute little useful information are therefore penalised towards zero and removed from the final model, while stronger predictors retain larger coefficient magnitudes, reducing overfitting while also limiting the number of retained features. Algorithmically, `glmnet` achieves this through coordinate descent optimisation, as described by Friedman et al. (2010), where the model iteratively cycles through each coefficient, updating one coefficient at a time while holding all others fixed until convergence is achieved. This approach handles the non-differentiability introduced by the L1 penalty through soft-thresholding and scales efficiently to datasets with large numbers of predictors. This is especially important when incorporating interaction-expanded feature as used in this model.

The prediction process produced by `glmnet_interact` remains fully interpretable and mathematically transparent, as predictions can be directly reconstructed from the final coefficient table shown in **Appendix 01**. The general prediction equation used to calculate the final response is given below:

$$\hat{y} = \beta_0 + \sum_{j=1}^p \beta_j x_j$$

where β_0 represents the intercept term and each β_j coefficient quantifies the contribution of predictor x_j to the final prediction. The intercept value of 3089.82 represents the baseline predicted response, while each coefficient represents the expected additive change associated with that predictor when all other variables are held constant. Larger absolute coefficient magnitudes therefore indicate predictors exerting stronger influence on the predicted outcome.

Several predictors retained relatively large coefficients within the final `glmnet_interact` model, suggesting substantial influence on the response variable. For example, *Exercise* produced the largest negative coefficient (-543.19), indicating that higher exercise levels were associated with considerably lower predicted response values. Conversely, *Alcohol* (352.24) and *Coffee* (311.53) produced strong positive contributions to the predicted outcome. Several reagent and blood-type interaction terms also retained comparatively large coefficients, such as *ReagentF × BloodType B* (-315.73), suggesting that the effect of certain reagents differed meaningfully depending on patient blood-type

characteristics. Similarly, *Exercise × ChemoTreatments* (−35.88) indicates that the relationship between exercise and the response changed for patients also receiving chemotherapy treatment.

Within a medical modelling context, this transparency is particularly valuable. The `glmnet_interact` model provides a clear and auditable relationship between predictors and predictions, allowing the contribution of individual variables and interaction effects to be directly inspected. Although the optimised `brnn_optim` model achieved slightly superior predictive accuracy, the transparent coefficient structure of `glmnet_interact` provides substantially greater interpretability and explainability, making it a strong alternative where understanding model behaviour is an important requirement.

Part F: Conclusion

Overall, this project demonstrated the value of systematic model comparison and candidate model selection within a predictive modelling framework. The use of the `caret` package provided a structured and reproducible approach to model training, hyperparameter tuning, and resampling-based performance evaluation, enabling fair comparison across a range of regression techniques. Through this process, it became evident that no single modelling approach is universally optimal; instead, model performance depends on the balance between flexibility, generalisation, and interpretability.

This framework enabled consistent comparison across multiple `caret`-based models, supporting the identification of an optimal performing model of `brnn_optim` in terms of predictive accuracy, alongside a more transparent alternative `glmnet_interact` that offered greater interpretability. The analysis highlighted the importance of considering both performance and explainability when selecting models for applied problems, particularly in healthcare contexts where model transparency is critical.

Ultimately, the study reinforces how structured tuning and evaluation within `caret` supports robust and informed model selection, ensuring that the final choice is both statistically reliable and appropriately aligned with the requirements of the problem.

Part G: References

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Part H: Artificial Intelligence Declaration

During the preparation of this report, DeepSeek, Claude and ChatGPT were employed to support code development, identify and resolve coding errors, and refine the written expression of the technical content. These tools were used solely for debugging assistance and stylistic improvements. The underlying data analysis, interpretation of results, and all substantive conclusions are entirely my own.

Part I: Appendix

Appendix 01: Coefficient values for glmnet_interact		
Coefficient	Coefficient Value	Type
main count	154	Main Effects
interaction count	0	Interactions
total count	154	Total Features Selected
(Intercept)	3089.82	Intercept
Exercise	-543.19	Main Effect
Alcohol	352.24	Main Effect
ReagentF x BloodType_B	-315.73	Main Effect
Coffee	311.53	Main Effect
BloodType_AB x BloodType_O	201.4	Main Effect
ReagentJ	166.24	Main Effect
ReagentB x BloodType_AB	-142.3	Main Effect
ReagentG	-140.63	Main Effect
ReagentK	-110.45	Main Effect
ReagentC	-86.87	Main Effect
ReagentH x BloodType_AB	-85.62	Main Effect
ReagentK x BloodType_O	-85.57	Main Effect
Exercise x ObservationDate_dow_Fri	84.69	Main Effect
ObservationDate_dow_Tue x ObservationDate_dow_Fri	84.12	Main Effect
ReagentB x BloodType_B	-78.78	Main Effect
ObservationDate_dow_Mon	76.71	Main Effect
Exercise x ObservationDate_dow_Mon	-75.84	Main Effect
ReagentF	72.94	Main Effect
ChemoTreatments	64.59	Main Effect
Alcohol x ObservationDate_dow_Fri	-54.87	Main Effect
ReagentF x ObservationDate_dow_Fri	-49.38	Main Effect
ReagentK x BloodType_B	-48.5	Main Effect
BloodType_AB	-47.56	Main Effect
ReagentF x BloodType_O	47.39	Main Effect
ReagentI x BloodType_B	46.42	Main Effect
ReagentG x BloodType_AB	46.26	Main Effect
Alcohol x ObservationDate_dow_Mon	45.67	Main Effect
Coffee x ObservationDate_dow_Fri	-45.43	Main Effect
Exercise x ObservationDate_dow_Tue	43.67	Main Effect
Coffee x ObservationDate_dow_Mon	40.47	Main Effect
Exercise x ChemoTreatments	-35.88	Main Effect
ObservationDate_dow_Fri	-34.76	Main Effect
ReagentC x ObservationDate_dow_Mon	-34.62	Main Effect
ReagentG x BloodType_O	34.31	Main Effect
Alcohol x ChemoTreatments	33.2	Main Effect
ReagentC x ObservationDate_dow_Fri	32.1	Main Effect
ReagentN	28.64	Main Effect
BloodType_B x ObservationDate_dow_Mon	26.78	Main Effect
ChemoTreatments x ObservationDate_dow_Mon	26.08	Main Effect
ReagentN x ObservationDate_dow_Mon	25.98	Main Effect
BloodType_AB x ObservationDate_dow_Fri	24.62	Main Effect
ReagentD x BloodType_B	-24.08	Main Effect
Coffee x ObservationDate_dow_Tue	-23.12	Main Effect
ReagentH	22.77	Main Effect
BloodType_O x ObservationDate_dow_Fri	21.24	Main Effect
Alcohol x ObservationDate_dow_Tue	-20.98	Main Effect
Coffee x ChemoTreatments	19.14	Main Effect
ReagentH x ObservationDate_dow_Tue	-18.22	Main Effect
ReagentG x ObservationDate_dow_Tue	17.84	Main Effect
BloodType_AB x ObservationDate_dow_Mon	-17.79	Main Effect
ReagentK x ObservationDate_dow_Mon	-16.83	Main Effect
BloodType_O x ObservationDate_dow_Mon	-16.68	Main Effect
ReagentB x ObservationDate_dow_Fri	16.37	Main Effect
ReagentJ x BloodType_B	-16.17	Main Effect
ChemoTreatments x BloodType_AB	13.69	Main Effect
ReagentH x ObservationDate_dow_Mon	13.59	Main Effect
ChemoTreatments x BloodType_B	-13.47	Main Effect
ReagentC x ReagentM	12.19	Main Effect
Alcohol x ReagentE	-11.22	Main Effect
Exercise x ReagentH	-11.12	Main Effect
ChemoTreatments x ReagentA	-10.78	Main Effect

Exercise x ReagentM	9.79	Main Effect
ChemoTreatments x ReagentN	9.14	Main Effect
Alcohol x ReagentH	8.31	Main Effect
ObservationDate dow Mon x ObservationDate dow Thu	-8.24	Main Effect
BloodType O x ObservationDate dow Tue	8.23	Main Effect
Coffee x BloodType AB	-7.93	Main Effect
BloodType AB x ObservationDate dow Tue	7.68	Main Effect
ChemoTreatments x ReagentK	-7.53	Main Effect
Exercise x ReagentK	7.32	Main Effect
Alcohol x ObservationDate dow Thu	7.27	Main Effect
ChemoTreatments x ObservationDate dow Fri	-6.98	Main Effect
ReagentN x ObservationDate dow Fri	-6.91	Main Effect
BloodType B x ObservationDate dow Fri	6.36	Main Effect
ChemoTreatments x ReagentG	-6.16	Main Effect
Exercise x ObservationDate dow Thu	6	Main Effect
ReagentE x ReagentK	-5.56	Main Effect
ChemoTreatments x ReagentD	5.5	Main Effect
ReagentJ x ObservationDate dow Sat	-5.34	Main Effect
ReagentK x ObservationDate dow Tue	5.34	Main Effect
ReagentM x BloodType B	-5.16	Main Effect
ReagentE x ObservationDate dow Sat	-4.81	Main Effect
ObservationDate dow Mon x ObservationDate dow Sat	-4.73	Main Effect
Coffee x ReagentI	4.71	Main Effect
ReagentA x ReagentI	4.7	Main Effect
Coffee x BloodType B	-4.53	Main Effect
Alcohol x ObservationDate dow Sat	4.16	Main Effect
ReagentM x ObservationDate dow Mon	3.88	Main Effect
BloodType O x ObservationDate dow Sat	-3.83	Main Effect
Alcohol x BloodType B	-3.77	Main Effect
ReagentC x ObservationDate dow Wed	-3.77	Main Effect
ReagentN x BloodType AB	-3.57	Main Effect
BloodType B x ObservationDate dow Sat	3.45	Main Effect
BloodType B	3.21	Main Effect
ChemoTreatments x ReagentE	3.2	Main Effect
Alcohol x Coffee	2.95	Main Effect
Coffee x ReagentM	2.7	Main Effect
ObservationDate dow Mon x ObservationDate dow Wed	-2.56	Main Effect
ReagentE x ObservationDate dow Thu	-2.55	Main Effect
Exercise x ObservationDate dow Wed	2.51	Main Effect
ReagentE x ObservationDate dow Wed	-2.23	Main Effect
Coffee x ObservationDate dow Sat	-2.17	Main Effect
Coffee x ReagentH	-2.08	Main Effect
ReagentD x ReagentF	2.06	Main Effect
Coffee x ObservationDate dow Thu	-2.02	Main Effect
ReagentF x ReagentL	1.98	Main Effect
ReagentH x ObservationDate dow Wed	-1.94	Main Effect
ReagentE x ObservationDate dow Fri	1.91	Main Effect
ChemoTreatments x BloodType O	1.89	Main Effect
Exercise x ReagentD	-1.84	Main Effect
ReagentG x ObservationDate dow Sat	-1.8	Main Effect
Coffee x ObservationDate dow Wed	-1.77	Main Effect
Exercise x ReagentG	1.76	Main Effect
ReagentC x BloodType O	1.69	Main Effect
ReagentE x ReagentM	-1.6	Main Effect
Exercise x BloodType O	-1.54	Main Effect
ChemoTreatments x ReagentM	1.45	Main Effect
Exercise x BloodType B	-1.39	Main Effect
ReagentE x BloodType AB	-1.13	Main Effect
Coffee x ReagentB	-1.07	Main Effect
ReagentF x BloodType AB	-1.04	Main Effect
ReagentB x ReagentI	1.02	Main Effect
Exercise x ReagentJ	-0.98	Main Effect
ReagentB x ReagentL	-0.96	Main Effect
ReagentK x ObservationDate dow Thu	-0.92	Main Effect
Exercise x ReagentN	-0.92	Main Effect
ReagentE x BloodType O	-0.88	Main Effect
BloodType B x ObservationDate dow Wed	-0.88	Main Effect
Coffee x BloodType O	-0.86	Main Effect
ReagentG x ObservationDate dow Wed	-0.76	Main Effect
Alcohol x ReagentJ	0.71	Main Effect
ReagentB x ReagentK	-0.68	Main Effect

ChemoTreatments x ObservationDate dow_Thu	0.6	Main Effect
ReagentA x BloodType B	0.58	Main Effect
Coffee x Exercise	-0.56	Main Effect
ReagentE x ReagentN	-0.43	Main Effect
ReagentG x ReagentN	0.41	Main Effect
BloodType_O x ObservationDate dow_Thu	0.39	Main Effect
ReagentA x ReagentN	0.3	Main Effect
ReagentD x ReagentG	0.28	Main Effect
ReagentC x ReagentE	0.28	Main Effect
ReagentC x BloodType AB	0.25	Main Effect
Alcohol x BloodType AB	0.24	Main Effect
ReagentA x ReagentD	0.23	Main Effect
ReagentM x ObservationDate dow_Wed	-0.18	Main Effect
Exercise x ReagentC	0.15	Main Effect
Exercise x BloodType AB	-0.09	Main Effect
ReagentB x BloodType_O	-0.08	Main Effect
ReagentK x BloodType AB	-0.05	Main Effect
ReagentG x ObservationDate dow_Mon	-0.05	Main Effect
Alcohol x BloodType_O	-0.05	Main Effect
ReagentL x ObservationDate dow_Fri	-0.02	Main Effect
ChemoTreatments x ObservationDate dow_Sat	0.01	Main Effect
ReagentD x ReagentL	0	Main Effect

Appendix 01: Full list of coefficients and coefficient values used within the glmnet_interact model. Due to the way the model is designed, it allows for very transparent communication of the calculations carried out by the model, in a highly interpretable manner. This transparency is particularly important for a model that simulates a medical response.